

Legends of SCIENCE

Book 4

Anne Clark



Legends of **SCIENCE**

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Anne Clark

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Introduction

People have questioned the world around them for thousands of years, leading to an accumulation of scientific knowledge through the efforts of many. As Isaac Newton said, 'If I have seen further, it is by standing on the shoulders of giants.'

This series uses the work of individual scientists to give students a context in which to explore ideas from many branches of science, including biology, chemistry, geology and physics. Each worksheet explores the life and work of an eminent scientist, including those living and dead, female and male.

Teachers will find these worksheets useful for extension, revision, homework and assessment.

They will also engage student interest in the field of science, providing a solid base for students to explore further both the contributions of individual scientists and the concepts they investigated. Some activities require access to research materials to complete the tasks.

The text of this book is also on the CD-ROM attached to the inside back cover.

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Rachel Carson

1907–1964

- Rachel Carson was born in Pennsylvania USA in 1907.
- Rachel wrote the book **Silent Spring** while she was dying of breast cancer. The book described the damaging effects of chemicals on the environment and led to DDT being banned in 1972 in the USA. It helped to develop an environmental consciousness worldwide.
- In 1980, Rachel was awarded the Presidential Medal of Freedom, the highest civilian honour in the USA.
- In 1999, Times magazine voted her one of the top 20 most influential scientists and thinkers of the 20th century.

DDT

1 Complete the sentences using these words: *insects, pesticide, contaminated, malaria, death, cancer, bees, ingestion, typhus, odourless, muscle, resistance, chains.*

Dichloro-diphenyl, trichlorethane (DDT) is an organochloride _____ invented in the 19th century. DDT controlled _____ that affected agricultural production and mosquitoes that transmitted _____ and _____. Entire cities were sprayed with DDT after World War 2 to kill fleas, lice, mosquitoes and flies.

DDT is a tasteless, _____, white, crystalline substance. It is absorbed by contact or _____ and causes plasma membranes to leak and release sodium and potassium ions. Neurone function is affected and this can result in _____ contraction, seizures, paralysis and _____.

Insects developed _____ to DDT and higher levels had to be used to kill them. DDT was not very biodegradable, and it dispersed worldwide. Penguins in Antarctica were _____ with DDT, even though it had not been used within 2000 km.

DDT killed beneficial insects such as _____ and accumulated in food _____. The number of falcons and eagles rapidly declined because DDT made their eggshells brittle.

DDT has been detected in human breast milk and is known to affect hormone and liver function and cause _____.

Definitions

- DDT** – Poison used to kill insects
- Food chain** – The feeding relationships in a community
- Biomagnification** – The accumulation of a chemical in a food chain
- Malaria** – Lethal disease transmitted by mosquitoes
- Lethal dose 50** – Amount of chemical required to kill 50% of a population

Characteristics of DDT

Refer to table 1 to answer these questions.

- 2 What elements does DDT contain?
- 3 Describe the appearance of DDT and explain why this may make the substance dangerous to living things.
- 4 For how long can DDT remain in the soil? What factors could determine how long it persists?
- 5 Where is DDT stored in the body?
- 6 Explain what the abbreviated units of measurement mean.
- 7 In your own words and using examples, describe how DDT could be transported to different places.

Table 1 – Characteristics of DDT

Name	DDT
Chemical formula	C ₁₄ H ₉ Cl ₅
Appearance	White crystalline chemical
Odour	Odourless
Taste	Tasteless
Melting point	109°C
Persistence in soils	1–15 years – longer in some regions
Absorption	Stored in fat cells
Method of transportation	Absorbed onto solid particles and transported by airborne particles and running water
Maximum residue level in food	0.02 mg/kg milk fat and 5 mg/kg meat fat*
Maximum permissible DDT residue levels in drinking water	1.0 µg/L*

* Source: WHO (World Health Organisation)

Environmental effects

Biomagnification

Table 2 shows the biomagnification (accumulation) of DDT in a food chain. A lake was sprayed to kill midges, and within a year, bird numbers declined.

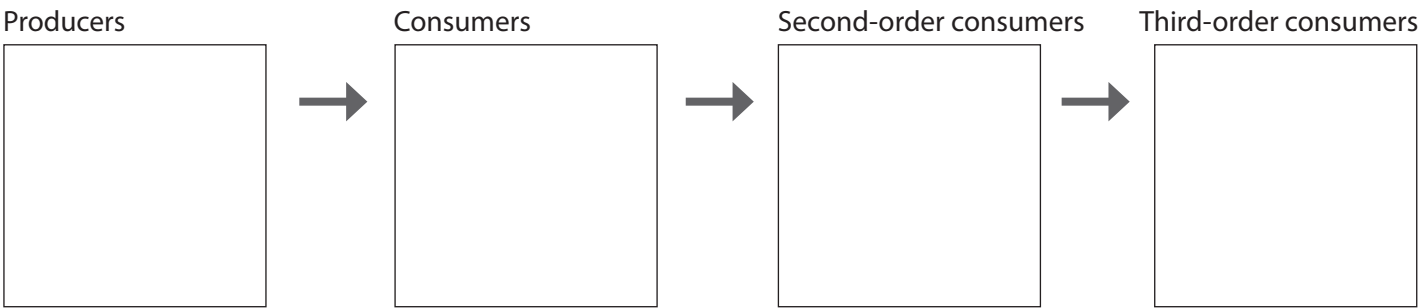
- 8 Draw a bar graph of the information in table 2.
- 9 How did DDT affect the birds at the lake?
- 10 Why do the lake trout have higher levels of DDT than the phytoplankton and zooplankton?
- 11 Complete figure 1, a food chain to show the accumulation of DDT in Lake Clear.

Table 2 – DDT levels in a food chain

Phytoplankton	0.0025 ppm
Zooplankton	0.123 ppm
Lake trout	4.63 ppm
Birds	124 ppm

Source: Environment Canada 1991, *A primer on water quality, questions and answers*. Environment Canada, Conservation and protection p.67

Figure 1 – Food chain of Lake Clear in California



- 12 Discuss the effects on an ecosystem when DDT kills bees and other beneficial insects.

Lethal doses

Table 3 indicates the lethal dose level of DDT for different organisms.

- 13 Which animal required the smallest lethal dose of DDT? Suggest reasons for this.
- 14 How much DDT is required to kill a dog?
- 15 Outline how humans could ingest DDT without knowing it.

Table 3 – Lethal doses of DDT for different organisms

Animal	Lethal dose to kill 50% of the population
Dog	150–750 mg/kg oral
Human	500 mg/kg oral
Fish	8–100 µg/L
House fly	0.033 µg/fly
Cat	150–600 mg/kg oral

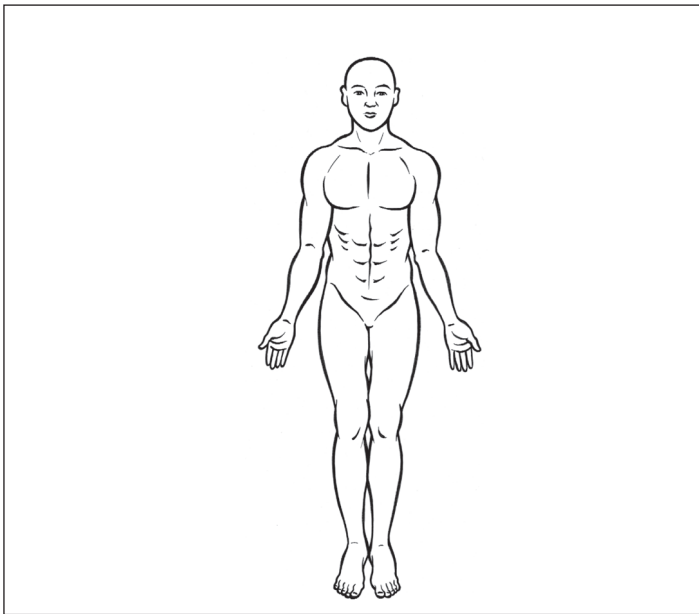
Symptoms of DDT poisoning

- 16 Use table 4 to help you label figure 2 to show where symptoms of DDT poisoning would occur in a human.

Table 4 – Symptoms of DDT poisoning

Vomiting	Dizziness
Tremors	Headaches
Changes in liver function	
Contaminated breast milk	

Figure 2 – Symptoms of DDT poisoning



Extension

- * In 1965, the National Academy of Sciences stated that in two decades, DDT had prevented 500 million human deaths that would otherwise have been inevitable. Discuss this, debating the effects of DDT.
- * Discuss the main uses of chemicals in our society. Outline their benefits and the problems they can cause. Consider poisons and medicines in your response.



Alec Jeffreys

1950–

DNA

DNA (deoxyribonucleic acid) is found in the chromosomes of the cell nucleus.

DNA is a double-helix molecule, consisting of two strands of Deoxyribose sugar, phosphates, and the organic bases adenine, guanine, cytosine and thymine.

The number and sequence of base pairs on the DNA is coded information which acts as the genetic blueprint for many characteristics, including skin, hair and eye colour. Except for identical twins, no two people have the same DNA.

- 1 Label the structure of DNA in figure 1 using these words: *adenine, guanine, cytosine, thymine, double helix*.

Uses of DNA fingerprinting

- 2 Complete table 1 by outlining how DNA fingerprinting could be used in each situation.

Table 1 – Uses of DNA fingerprinting

SITUATION	USES OF DNA FINGERPRINTING
Immigration disputes	
Paternity or maternity disputes	
Law enforcement	
Forensics	
Wildlife population genetic studies	
Identifying victims of war	
Identifying inherited diseases and developing therapies	
Archaeology, e.g. joining fragments of the Dead Sea Scrolls	

Crime detection and DNA fingerprinting

- 3 Complete the sentences using these words: *DNA, criminal, 18, prison, Nazi, died, murder, testing*.
- a** In 1987, British baker Colin Pitchfork became the first _____ caught using DNA fingerprinting.
- b** In 1989, Chicago man Gary Dotson became the first person to have their criminal conviction overturned because of _____ evidence.
- c** In 1992, DNA evidence was used to show that Josef Mengele, a criminal _____ doctor, had _____ and was buried in Brazil under the name of Wolfgang Gerhard.
- d** In 1994, detectives used fur samples from a cat named Snowball to link a man to the _____ of his wife.

- Alec Jeffreys was born in Bedfordshire, England. After graduating from Oxford University, he undertook genetic research at Leicester University.
- In 1984 he developed DNA fingerprinting (profiling) – a method used to identify a person by their DNA.
- Today, DNA fingerprinting is used worldwide in fields such as medicine, law enforcement and archaeology.
- Alec was knighted in 1994 and has been awarded the Albert Einstein World Award for Science and the Louis Jeantet Prize for Medicine.

Figure 1 – Structure of DNA



Definitions

DNA – Deoxyribonucleic acid, a chemical containing the genetic material of an organism

Paternity testing – Identifying the father of a child through DNA testing

Forensic science – The application of science to criminal investigations

- e In 2003, new DNA evidence resulted in three murderers, Dennis Halstead, John Kogut and John Restivo, being retried for their crimes. These men had already served ____ years of their 30 year sentences.
- f In 2005, Robert Clark was proved innocent of attacking a woman after serving 24 years in _____. He is the 164th person in the United States to be freed due to post-conviction DNA _____.

Reliability of DNA fingerprinting

- 4 Complete table 2 by explaining how each factor could affect the reliability of DNA fingerprinting.
- 5 Should DNA fingerprinting be relied upon as the only evidence in convicting a criminal? Explain your answer.

Table 2 – Factors affecting DNA fingerprinting

FACTOR	EFFECT
Contamination by police at the scene of a crime	
Incorrect laboratory techniques	
Deliberate alteration of DNA at the crime scene, e.g. planting DNA	
Age of evidence, e.g. old murders	

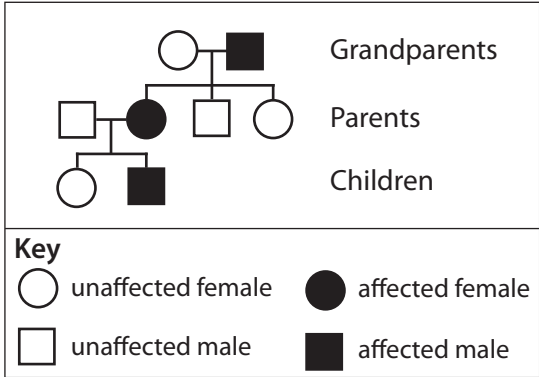
Medical uses of DNA fingerprinting

Muscular dystrophy

Muscular dystrophy is an inherited disease, characterised by the progressive wasting of the skeletal muscles.

- 6 Study figure 2 at the right.
- a Who was the first person in this family to have the disease?
- b How many males were affected by the condition?
- c How many generations were affected by the disease?
- d How do you think the next generation will be affected?
- 7 Draw up a table and outline the arguments for and against having children when DNA fingerprinting shows the presence of the muscular dystrophy gene.

Figure 2 – Muscular dystrophy in a family tree



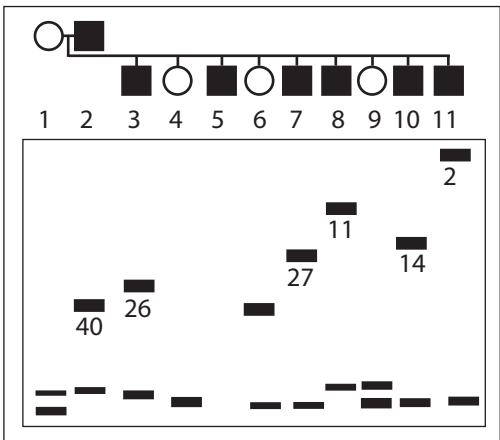
Huntington's disease

Huntington's disease is a degenerative brain disease caused by a mutation on chromosome four. Approximately one in 30 000 people have the disease. A parent with the Huntington gene has a 50% chance of passing this disease onto their children.

The disease destroys all the cells in the basal ganglia, the part of the brain that contains movement, emotion and cognitive ability. Affected people normally die at 40–50 years of age. Symptoms include poor memory, depression, lack of coordination, difficulty walking and speaking, slurred speech and intoxicated appearance.

- 8 Figure 3 shows the pedigree of a family affected by Huntington's disease.
- a What is the probability of a child inheriting this disease?
- b If a child doesn't receive the gene for the disease, can it still affect their children?
- c Which children have been affected by the disease?
- 9 Some banks and insurance companies have refused mortgages, health and life insurance to families with Huntington's disease. Outline the moral and ethical arguments of this decision.

Figure 3 – DNA fingerprint of a family with Huntington's disease



Extension

- * Choose one specific application of DNA fingerprinting and explain how it has made the world a better place.
- * *The world would be safer if everyone's DNA was catalogued at birth, so it could be checked any time a crime was committed.* Do you agree? Justify your answer.
- * *Inherited diseases could be eliminated from the world with the use of DNA fingerprinting.* Discuss this statement.



Karl Jansky

1905–1950

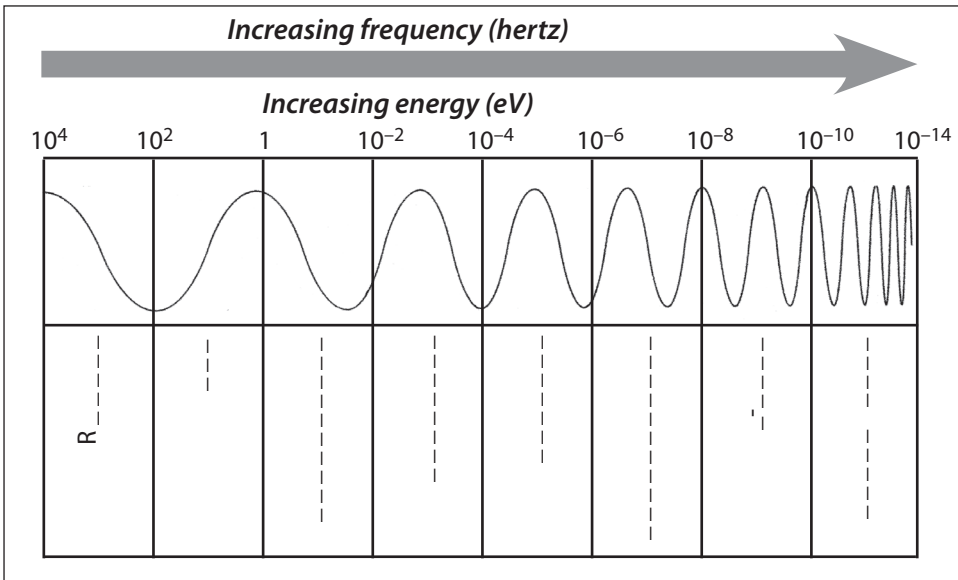
- Karl Jansky was born in Oklahoma, USA. He attended the University of Wisconsin and received a Bachelor of Science in Physics in 1927.
- In 1928 he worked with Bell telephone laboratories and built a radio antenna to find the cause of static on long distance telephone calls. Karl found that the static was caused by radio waves from the centre of the Milky Way.
- Radio astronomers use the term Jansky as the unit of measurement for the strength of radio signals.

Structure of radio waves

Light, X-rays, microwaves, infrared and radio waves are all electromagnetic waves with different wavelengths and frequencies.

- 1 Label the electromagnetic spectrum in figure 1 using these words: *Radio, UHF, Microwaves, Infrared, Visible, Ultraviolet, X-rays, Gamma rays.*

Figure 1 – The electromagnetic spectrum



Definitions

Radio wave – Electromagnetic wave

Pulsar – Rotating neutron star

Radio telescope – Device used to detect radio waves from outer space

Frequency of different waves

- 2 Use figure 1 to state the wavelength of these waves.
 - a Ultraviolet
 - b Radio
 - c X-rays
 - d Light

Sound and radio waves

Sound waves and radio waves are not the same. Sound is a longitudinal wave and cannot travel through a vacuum. Radio waves are electromagnetic waves and can travel through a vacuum.

When you turn on a radio, the transmitter at the radio station converts the sound into an electrical signal and encodes it onto a high frequency radio wave. The radio receiver decodes the information and the speaker converts the electrical signal back into sound.

- 3 What is the difference between sound waves and radio waves?

Radio signals and the universe

- 4 Complete the sentences using these words: *distances, supernovas, radio, flares, Bang, radiation, magnetic, Milky, pulsars, temperatures.*

_____ signals from outer space have been used to measure _____ in space and determine the structure of the _____ Way.

These signals have found _____ and quasars, provided evidence for the Big _____ theory and helped determine the _____ of planets.

Sources of radio waves from outer space include thermal radiation from solid bodies such as planets, thermal _____ from hot gas in the interstellar medium, electrons in weak _____ fields, spectral line radiation from gases around stars, radiation from neutron stars, solar _____ and sunspots, cosmic background radiation and remnants of _____ or exploding stars.



Russell Ohl

1898–1987

- Russell Ohl was born in Allentown, Pennsylvania. When he was 16, he went to Pennsylvania State College and undertook a degree in electrical engineering.
- During World War 1 he served in the Army Signal Corps and afterwards worked at Bell Laboratories. He worked on equipment for radio and military radar.
- Ohl found that germanium could be used to make a semiconductor material for diodes. This led to the development of LEDS and laser diodes.
- In 1941, Ohl developed the silicon solar cell which is now used worldwide.

History of solar energy

1 Complete the sentences using these words: Sun, coal, 10, current, energy, solar, wave, silicon, ancient, metal, weather, less.

The _____ is a star and each year it provides _____ times more energy than could be produced from all of the world's reserves of _____ and oil.

The sun provides _____ through heat and photosynthesis. Fossil fuels consist of _____ energy that has been trapped in plant remains. The Sun also controls the Earth's _____ systems thereby creating wind and _____ power. The Greeks and Romans used passive solar-powered houses in _____ times.

In 1839, Antoine Cesar Becquerel observed the photovoltaic effect. He found that shining light on certain _____ electrodes in a conductive solution created an electric _____.

In 1877, Charles Fritts developed a solar cell made from selenium. However, it was _____ than 1% efficient.

In 1941, Russell Ohl developed the first _____ solar _____ cell and these are now widely used in homes, industry, space and telecommunications.

Definitions

Solar cell – Device used to generate energy from the sun
Silicon – Non-metallic element used in solar cells

Passive solar-powered house – A house facing north that has large windows and tiles and is heated by the sun

Table 1 – Uses of solar energy

Situation	Example of use of solar cell
Telecommunications	Energy source in isolated regions
Oil companies	
Highway safety equipment	
Public telephone boxes	
Roadside signs	
Passive solar powered houses	
Satellites	
Calculators	
Watches	
Electricity for remote locations	
Water pumps	
Refrigeration	
Caravan and boating industries	
Space craft	

Solar cells and efficiency levels

A solar cell consists of two layers of the semiconductor silicon. Impurities are added to the two layers to alter their chemical properties. One layer has more negatively charged electrons (n type) and the other layer has less (p type). When light strikes the cell, electrons move across the junction and this creates an electric current.

2 Describe the trends in solar cell usage and suggest reasons for these trends.

Uses of solar energy

3 Complete table 1 by listing an example of the use of solar cells in each situation. The first one has been done for you.

Advantages and disadvantages of solar energy

- 4 Identify each dot point as an advantage of solar energy (A) or as a disadvantage (D).
- * Cells are expensive ____

* Cells are inefficient ____

* Renewable energy source ____

* Needs sunshine to operate ____

* Clean ____

* Doesn't work at night time ____

* Non-polluting ____

* Individual users can generate their own power without large scale power stations ____

* No moving parts to break ____

* Low maintenance ____

* Limitless energy ____

* Free energy ____

* Flexible to growing needs e.g. each new roof can have rooftop power ____

* Unlike fossil fuels, does not release carbon dioxide ____
- World energy production
- 5 Study graph 1, which shows the world production of primary energy.

a Calculate the percentage for each energy source.

b What percentage of total energy is provided by geothermal, solar, wind and wood?

c What percentage of total energy is derived from renewable sources?

d What percentage of total energy is derived from non-renewable sources?

6 What are the negative effects of using fossil fuels?

7 What government initiatives do you think would encourage the use of solar energy?

8 What technological developments do you think are needed to increase the use of solar energy?
- Graph 1 – World production of primary energy in quadrillion units
- | Energy Source | Production (quadrillion units) |
|----------------------------------|--------------------------------|
| Petroleum | 149.7 |
| Coal | 84.90 |
| Natural gas | 87.31 |
| Hydroelectric | 27.10 |
| Nuclear | 25.25 |
| Geothermal, solar, wind and wood | 28.3 |
- Source: US DOE EIA
- Factors affecting solar energy use
- 9 Outline how each factor in table 2 could promote or hinder the increased use of solar energy.
- Table 2 – Factors affecting solar energy use
- | Factor | | Effect on the use of solar energy | |
|--|--|-----------------------------------|--|
| Price of land – solar panels require large areas of land | | | |
| Cost of mining sand to make silicon cells | | | |
| Availability of sunshine | | | |
| Maintenance requirements | | | |
| Isolation, e.g. islands | | | |
| Conversion efficiency | | | |
| Education | | | |
| Cost of alternative energy sources | | | |
| Government incentive programs | | | |
- True or false?
- 10 Mark each statement as true (T) or false (F) and explain your reasoning.

a Solar cells are very efficient. ____

b Petroleum is the world's major source of energy. ____

c Solar energy contributes to the greenhouse effect, global warming and the creation of acid rain. ____

d Solar energy is a renewable source of energy. ____

e Photosynthesis is the conversion of solar energy into chemical energy. ____
- Extension
- * Discuss the disadvantages and advantages of our major energy sources.

* Research a recent technological development in a renewable energy source, such as solar, wind, wave or geothermal energy.

* List 10 ways you can reduce energy consumption at home or at school.
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11



Alexander Parkes

1813–1890

- Alexander Parkes was born in England. He was an apprentice to a brass founder, then became the manager of a casting department.
- In 1862 he developed the plastic Parkesine from cellulose treated with a solvent and nitric acid. This plastic was similar to rubber and could be carved into different shapes.
- In 1866, Alexander established the Parkesine Company and tried to mass produce the plastic. However, high costs and competition from natural rubber led to its closure in 1868.

Plastics

- 1 Complete the sentences using these words: *today, tonnes, consumption, natural, materials, plastic, carbon, 20, modified, polyethene.*

The word _____ is derived from the Greek word *plastikos*, meaning 'to mould and form'. It is used to describe _____ which have plastic properties.

The world's annual _____ of plastic materials has increased from approximately five million _____ in the 1950s to nearly 100 million tonnes _____. We produce and use _____ times more plastic today than we did fifty years ago.

Most plastics are made of polymers of _____ with oxygen, nitrogen, chlorine or sulphur.

Plastics were originally made from _____ materials such as shellac. They can be made of chemically _____ natural materials such as natural rubber and nitrocellulose, or artificially made from petroleum, for example _____.

Definitions

Plastics – Diversity of materials made from natural materials or chemicals such as petroleum, e.g. nylon

Biodegradable – Will decompose or break down

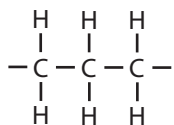
Bioplastic – Biodegradable plastics

Fauna – Animals

Chemical structure of plastics

- 2 Figure 1 shows the chemical structure of four common plastics. Research three uses of each plastic and list them below the chemical structures.

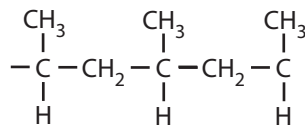
Figure 1 – Chemical structure of some common plastics



Polyethene

Uses:

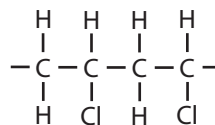
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Polypropylene

Uses:

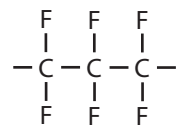
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PVC

Uses:

-
-
-



Teflon

Uses:

-
-
-

- 3 List the elements present in each plastic in figure 1.

Products made from plastic

- 4 List 10 items made with plastic that you use every day.
- 5 List 10 items that are not made with plastic that you use every day.

Types of plastics and their uses

6 Write each plastic beside the correct uses in the table at the right: *bakelite, polystyrene, nylon (polyamide), polyethylene, polyester, polyvinyl chloride, polycarbonate, polypropylene*.

7 What characteristics of plastics make them so versatile and popular?

8 What problems are created by using plastics for packaging? How could these problems be resolved?

9 Australians use approximately 6 billion plastic bags each year. Around 3.3 billion are supermarket plastic bags. If there are 21 million people in Australia, calculate how many plastic bags are used on average by each person, each year.

10 Describe the strategies people are using to decrease the use of plastic shopping bags in your community.

11 Identify each dot point as an advantage of plastics (A) or as a disadvantage (D).

- * Versatile ____
 - * Made from non-renewable fossil fuels ____
 - * Lightweight ____
 - * Durable ____
 - * Can release toxic fumes when burnt ____
- * Resistant to chemicals, water and impact ____
 - * Safe and hygienic ____
 - * Thermal and electrical insulation ____
 - * Generally non-biodegradable ____
- * Inexpensive to produce ____
 - * Can hurt fauna, e.g. birds and fish ____
 - * Some can be recycled ____
 - * Fill landfill sites ____
 - * Create secondary industries ____

Recycling plastics

- 12 Why is reusing plastics preferable to recycling plastics?
- 13 List three examples of ways that plastic materials can be reused.
- 14 Suggest three ways of promoting the reduction, reuse and recycling of plastics.

Bioplastics

Some biodegradable plastics have been developed. They contain plant sugars. Genetic engineering is also being used to develop micro-organisms that can break down plastics.

15 Which problems with plastics could be resolved by the development of biodegradable plastics?

True or false?

- 16 Mark each statement as true (T) or false (F). Then correct the false statements so that they are true.
- a All plastics are non-biodegradable. ____

b Plastics can kill fauna. ____

c Plastics are made of polymers or repeating units. ____

d It is safe to burn plastics. ____

e Many plastics are derived from fossil fuels. ____

f Nylon is a common plastic. ____

g Plastics can be an environmental hazard. ____

Extension

- * What recycling initiatives are there in your region? Describe them and explain how effective you think they are.
- * *The recycling of materials uses more energy than it saves.* Discuss this statement.



Albert Einstein

1879–1955

- Albert Einstein was born in Germany. In 1900, he graduated from Zurich Polytechnic and then worked at the patent office in Berne, Switzerland.
- While working at the patent office, he obtained his PhD and published four major works.
- His discoveries include: Brownian motion, Photon theory of light, The equation $E = mc^2$, Special theory of relativity.
- In 1915, Einstein developed the General Theory of Relativity.
- In 1933, Einstein fled Germany and moved to America. In 1939, he wrote to President Roosevelt urging America to develop the atomic bomb before Germany did.
- Later, he tried to develop a theory which unified all the laws and forces in nature as a single fundamental equation.

Energy and mass

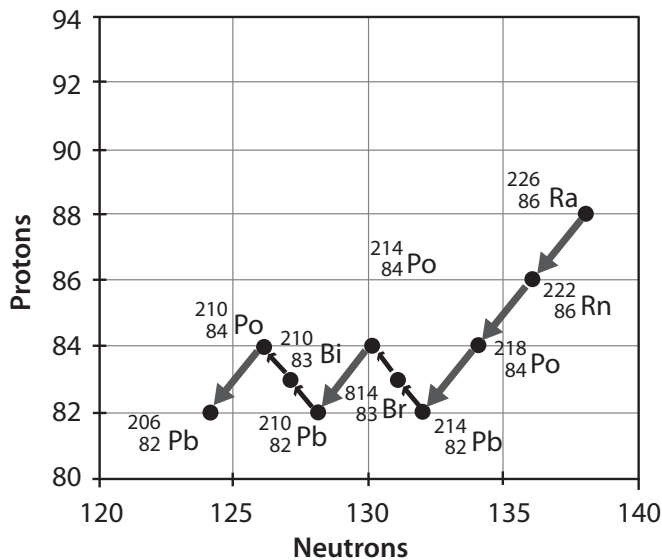
The equation $E = mc^2$ states that energy equals mass times the speed of light squared. This equation explains natural processes such as radioactive decay, carbon dating, energy generation in nuclear reactors and atomic bombs.

Radioactive decay

When radioactive material breaks down, it converts mass into energy in direct accordance with the equation $E = mc^2$.

Decay of radium

Figure 1 – Decay of radium



Definitions

Brownian motion – Zig-zag motion observed when a sample of pollen is put on a microscope slide. This movement is caused by the atoms in water bombarding the pollen

Radium – Radioactive element that decomposes to form lead

Uranium – Radioactive element used to produce energy in nuclear reactors

- Study figure 1, which shows the decay of the radioactive element, radium.
 - Name the element produced by the reaction.
 - Why do radioactive elements break down with time?
 - What is the source of the energy that is released by the reaction?

Carbon dating

Carbon-14 is used to determine the age of historic remains. Carbon-14 decays at a predictable rate, and so the amount left in an artefact can indicate its age.

- Study table 1, which shows the decay of carbon-14 with time.
 - Graph this data.
 - Describe and explain the shape of the graph you drew.
 - How many years does it take for half of the carbon to break down?

Table 1 – Decay of carbon-14 with time

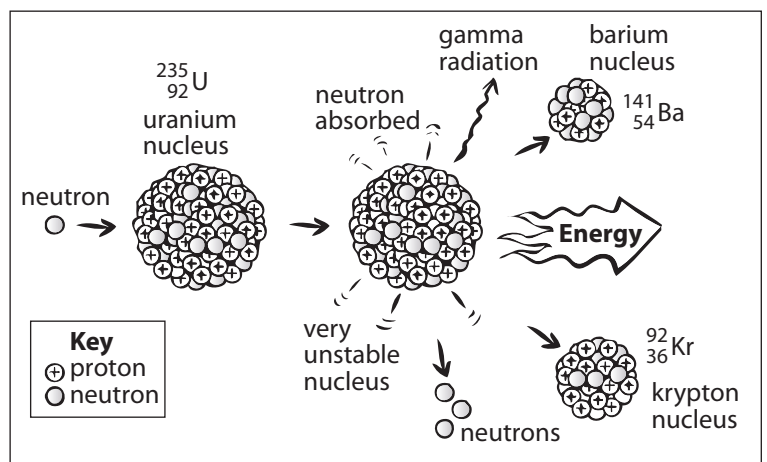
Time (years)	0	5 730	11 460	17 190	22 920
No. radioactive nuclei (%)	100	50	25	12.5	6.25

Nuclear reactors

Figure 2 shows the reaction used to produce energy in a nuclear reactor.

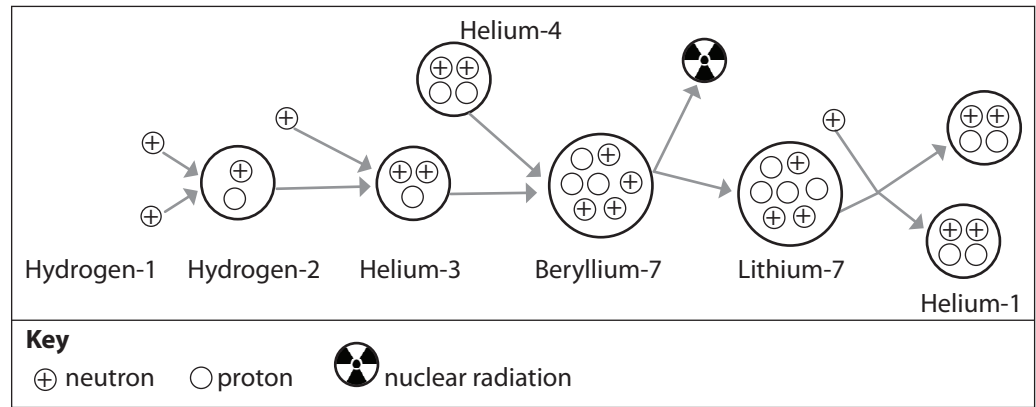
- Outline the process occurring in the reaction in figure 2.

Figure 2 – Reaction used to produce energy in a nuclear reactor



- 4 What particle is used to start the reaction?
- 5 What elements are created during the reaction?
- 6 What type of radiation is released during the reaction?
- 7 What is the source of the energy produced during the reaction?

Figure 3 – Nuclear fusion in the Sun



The Sun

The Sun produces energy by the process of nuclear fusion. In this reaction, hydrogen atoms are joined together to make helium and other elements.

- 8 Outline the process occurring in figure 3.
- 9 The mass of the helium nucleus produced is less than the mass of the original materials. Account for this 'loss' of mass.

Nuclear weapons

- 10 'The mushroom cloud from an atomic bomb explosion is $E = mc^2$ made visible.' Explain this statement.

Time

Einstein believed that a large mass could warp space and time. He believed clocks would slow down near a large gravitational mass like the Sun.

In 1971, two scientists flew four atomic clocks around the world. Their change in time due to speed and height showed that Einstein was correct.

A clock will speed up by 1.09 parts in 10^{13} for every kilometre above sea level. This is equal to 9.43 nanoseconds per day.

- 11 Study figure 4. Which clock will be faster? Why?

Bending of light by gravity

The curved space and time predicted by Einstein, and gravity's ability to bend light as it passes near a massive object, were first observed by Arthur Eddington in 1919.

The bending of starlight by gravity and a tiny shift in the orbit of the planet Mercury support Einstein's theories.

- 12 Referring to figure 5, explain why the actual location of a star is different to where the star appears to be.

True or false?

- 13 Mark each statement as true (T) or false (F). Then correct the false statements so that they are true.
 - a Energy and mass are not interchangeable. ____
 - b Time cannot change speed. ____
 - c Large objects can bend light. ____
 - d Gravitation and acceleration can be equal. ____

Figure 4 – Time in a valley and on a mountain

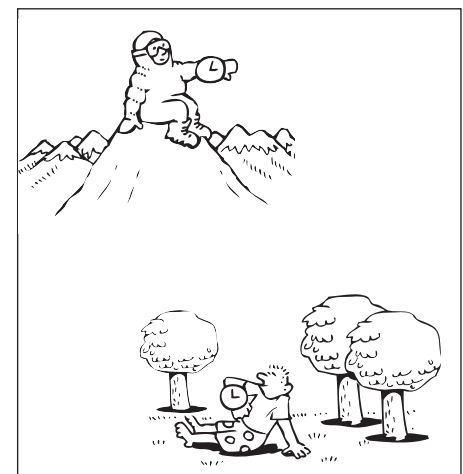
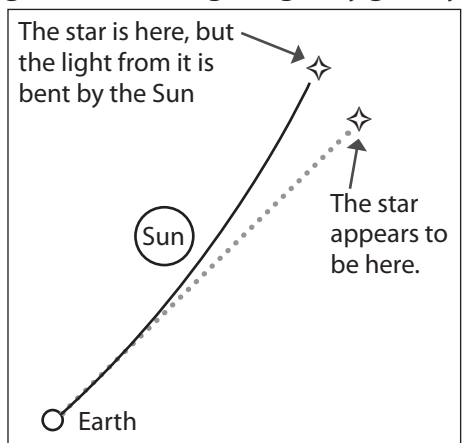


Figure 5 – Bending of light by gravity



Extension

- * Research nuclear power stations. Find out what materials are needed for the nuclear reaction and how much energy can be produced.
- * Should scientific discoveries be used to make weapons? Explain your answer.
- * Research one of Einstein's discoveries. Find out how it contributed to change in our society.



Annie Jean Macnamara

1899–1968

- Annie Jean Macnamara was born in Australia. In 1922 she obtained a Bachelor of Medicine from the University of Melbourne.
- In 1931 she showed that more than one strain of virus caused polio. This information helped create an effective vaccine against polio
- Annie used physiotherapy to treat polio, which helped patients to move their limbs.
- She campaigned for the use of the virus myxomatosis to control plague rabbit populations, which significantly increased sheep and wool production in Australia.

Polio (poliomyelitis)

1 Complete the paragraphs using these words: *nervous, paralysis, die, vaccine, iron, breathe, polio, immunisation, five, viral, water, 50, bloodstream.*

Polio is a _____ disease spread by ingesting faecally contaminated food and _____. More than _____ percent of cases occur in children between the ages of three and _____.

The virus infects the intestinal wall, moves via the _____ to the central _____ system and causes muscle weakness, _____ and death.

One in 200 infections leads to irreversible damage and paralysis. Of those paralysed, 5–10% _____ because their muscles are immobilised and they cannot _____. In 1928, Philip Drinker and Louis Shaw developed the _____ lung (a large metal tank equipped with a pump) that allowed polio victims to breathe.

In 1952, 60 000 people in the United States contracted _____ and 3000 died.

Mass immunisation programs of the Salk _____ were initiated in 1957. This was replaced by the Sabin oral vaccine (a live attenuated vaccine) in 1962 because it was easier to administer and provided longer lasting _____. By 1977, 254 000 people in the United States had been paralysed by polio.

Definitions

Polio – Viral disease spread by contaminated food and water
Vaccine – Dead or diluted form of disease given to a person to prevent that disease

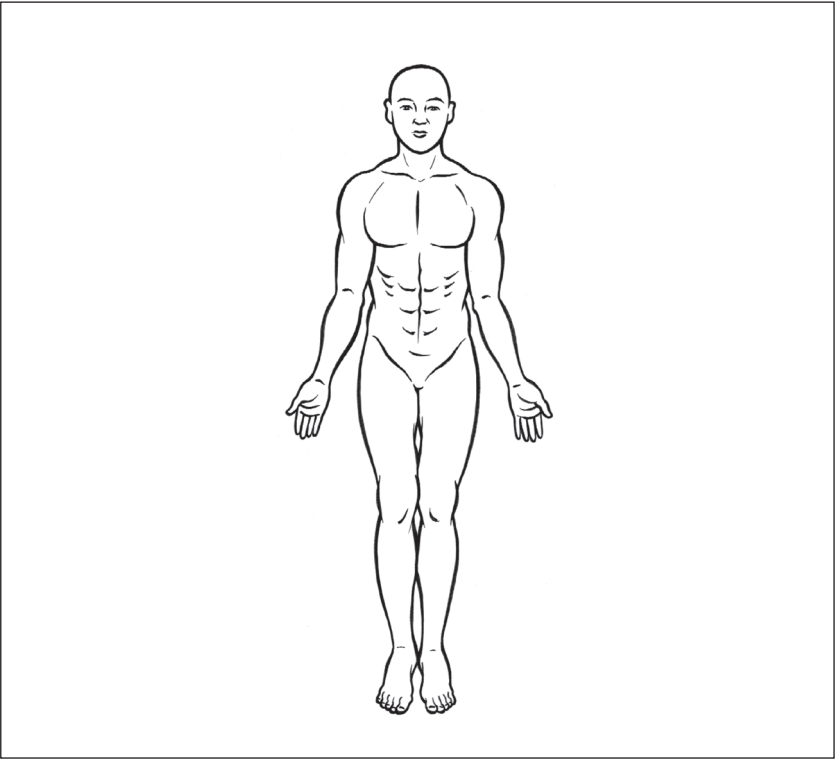
Symptoms of polio

2 Referring to the list of symptoms below, label figure 1 to show where the symptoms of polio occur on a person.

Symptoms of polio

* upper respiratory infection	* muscle weakness
* diarrhoea	* sensitivity to light
* fever	* neck and back stiffness
* sore throat	* pain in limbs
* general feeling of ill health	* paralysis of the arms and legs
* nausea	* paralysis of lungs causing respiratory difficulty
* vomiting	
* extreme fatigue	

Figure 1 – Symptoms of polio



Incidence of polio in the world

Polio is still prevalent in six countries: India, Nigeria, Egypt, Pakistan, Afghanistan and Somalia.

3 Describe where these countries are located in the world and what the living standards might be. Consider the socio-economic wealth of the communities and access to health facilities.

- 4 Explain why polio has not been eradicated in these regions.
- 5 How do you think food and water become contaminated with the polio virus?
- 6 Describe what you think would be the social effects of polio on a person who contracts the virus, their family and their community.

Treatment

Presently there is no cure for polio. The most effective way of controlling it is by eradication. In 1988, the World Health Organisation started the Global Polio Eradication Initiative. By 2001, more than 575 million children had been vaccinated in 94 countries.

Eradication efforts

Table 1 shows the number of cases of polio in the world from 1988 to 2005.

- 7 Graph this data. From your graph, what trends are apparent?
- 8 Has the eradication campaign been effective? Explain your answer.
- 9 How can polio recur in regions where it has been eradicated?
- 10 Suggest reasons why some parents do not have their children vaccinated.

True or false?

- 11 Decide whether each statement is true or false, and explain your reasoning.

a Contaminated water is one of the greatest causes of disease in the world.

b Polio can be prevented by vaccination.

c An iron lung allows a person to breathe.

d Polio cannot recur.

Table 1 – Cases of polio
1988–2005

Year	Number of cases of polio
1988	350 000
1996	4 074
1997	5 185
1998	6 349
1999	7 141
2000	2 971
2001	498
2002	1 922
2003	784
2004	1 258
2005	1 998

Source: World Health Organisation
Global Number of Poliomyelitis cases
www.who.int/healthinfo/statistics/maps

Other diseases caused by viruses

- 12 Complete table 2 by listing symptoms and control methods for each disease.

Table 2 – Diseases caused by viruses

Name	Symptoms	Method of control
Influenza		
Rabies		
Smallpox		
Mumps		
Hepatitis		
Distemper of dogs		
Foot and mouth disease		

Extension

- * List three viral diseases not mentioned on this worksheet. For each one, find out how it is spread, symptoms and how it is treated.
- * List five common vaccinations given in Australia. Briefly describe the history of each vaccination.
- * Education provides the basic mechanism for the effective control of polio. Discuss this statement.
- * Water quality and hygiene are two of the most important standards to be addressed in countries affected by disease. Discuss this statement.



Philo Farnsworth

1906–1971

- Philo Farnsworth was born near Beaver City, Utah, USA.
- In 1923, after two years of high school, Philo gained admission to Brigham Young University.
- In 1927, Philo created an image dissector camera tube. In 1929, he transmitted the first human images by television.
- Philo eventually sold his television patent to RCA Victor for \$1 million dollars.

How television works

- 1 Complete the sentences using these words:
fibre, microwave, data, light, electronic, higher, cathode, radio.
- Television stations relay sound and images using _____ waves, coaxial cable or _____ optics. The camera converts _____ waves into _____ impulses. The television receiver converts these impulses into a picture on the screen of a _____ ray tube.
- Traditional satellite television uses _____ frequencies to send images. New systems operate at a _____ frequency, and convert the images into digital _____ which is relayed through the satellite link.

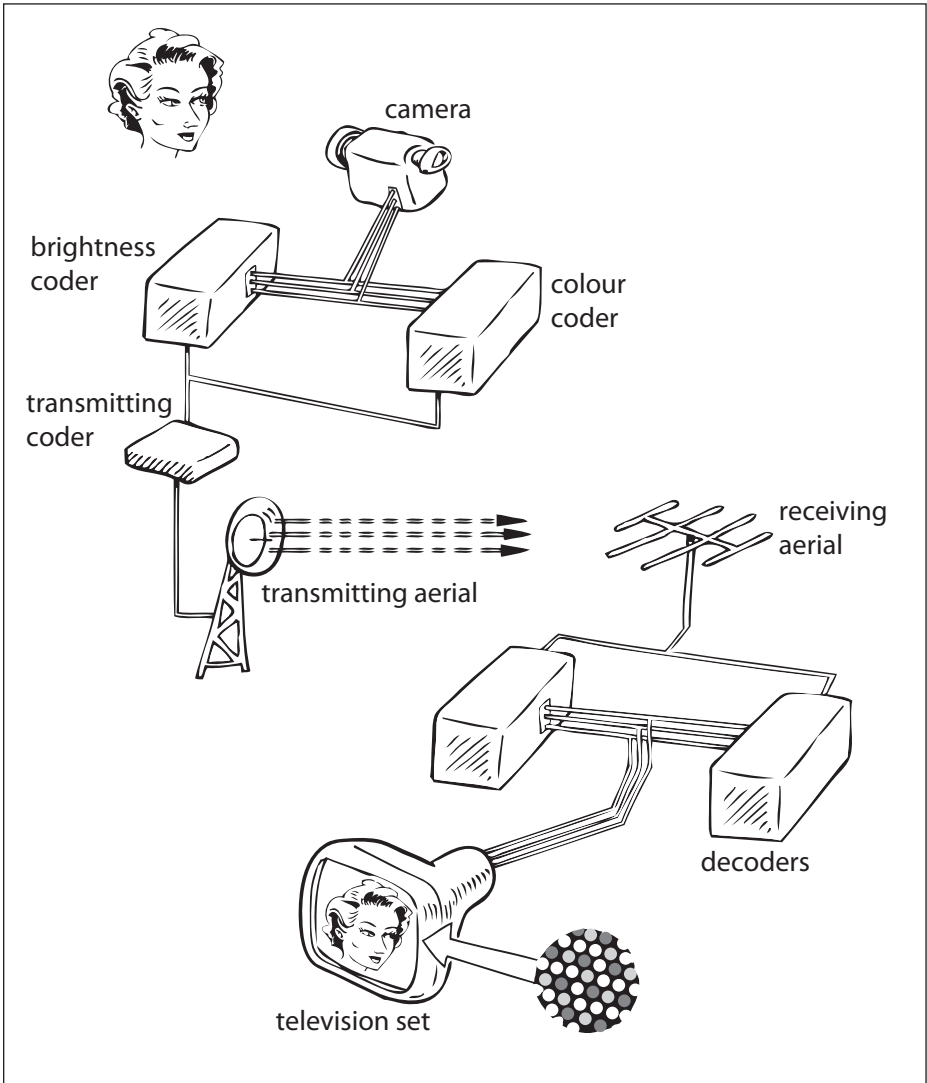
- 2 Referring to figure 1 and the above paragraph, describe how images are transmitted to your home.

Definitions

Radio waves – electromagnetic waves used to transmit television images

Patent – the legal ownership or rights of an invention

Figure 1 – Transmission of television images to your home



Use of television

Table 1 shows the percentage of Australian homes that owned a television from 1956–1993.

- 3 Graph the data in table 1.
- 4 Describe the trends shown in your graph, and suggest reasons for these trends.
- 5 Describe the public services that television programs provide today. For example, Play School educates young children.

Table 1 – Percentage of Australian homes that owned a television from 1956–1993

Year	1956	1961	1966	1971	1976	1981	1986	1991	1993
%	1	55	87	91	92	92	93	99	99

Source: Bureau of Transport and Communications Economics (1994) – Statistical summary of the Communications, Entertainment and Information industries.

Television in Australia

- 6 Write the correct date or time period beside each event in table 2: 2012, 1950s, 1966, 1960, 1971, 1978, 1986, 1994, 1993, October 1980, 20 July 1969, 9 December 1961, 22 November 1956, 1 January 2001, 16 September 1956, 15 June 1967, 1 March 1975.

Table 2 – Important events in the history of television in Australia

Date	Event
	Bruce Gyngell launched television in Australia with the words ‘Good evening, and welcome to television’.
	The first pay TV service became available.
	During this year, television broadcasting began in Darwin, Kalgoorlie and Mount Isa.
	Estimated date for Australia’s changeover to digital television.
	The first regional television broadcast began in Gippsland, Victoria.
	First live satellite transmission between Australia and the United Kingdom.
	By this year, about 70% of homes in Sydney and Melbourne had a television set, but regional television broadcasting didn’t occur until the following year.
	The Apollo 11 moon landing was televised live.
	Colour television arrived in Australia.
	By this year, over 64% of households in Melbourne and Sydney had colour television sets.
	SBS TV was launched in Sydney and Melbourne. It provided much international and foreign-language content, to cater for Australia’s growing multicultural population.
	AUSSAT, a new domestic satellite was introduced. This meant that television could be broadcast to more remote areas of Australia, without needing to set up new stations.
	Community television stations opened in Sydney, Melbourne, Brisbane, Adelaide and Perth.
	By the end of this decade, television had arrived in Brisbane, Adelaide and Perth. However, less than 5% of homes in Melbourne and less than 1% of homes in Sydney had television sets. At the time, a television set cost approximately 6–10 weeks’ wages.
	Digital television was launched in Australia.
	Five television stations in Melbourne and Sydney broadcast the Opening Ceremony of the Melbourne Summer Olympics.
	Test colour television transmissions were made for the first time.

- 7 Draw up a timeline to show the information in table 2. Then write a brief summary of what your timeline shows about the history of television in Australia.
- 8 Today, the average weekly wage in Australia is about \$1150. In the 1950s, a television set cost the equivalent of 6–10 weeks’ wages. Work out the total of 10 weeks’ average wages, then list three items that have an equivalent cost. What does this tell you about the cost of television sets in the 1950s?

Extension

- * Find out how LCD and plasma displays work.
- * Discuss how increasing delivery of television programs via the Internet may affect the future of television in the next 10 years.
- * Find a person who remembers life before television. Interview them about the changes they have seen in television, and how they feel these changes have contributed or reflected changes in society since the 1950s.



Willem Kolff

1911–

- Willem Kolff was born in Leyden, the Netherlands. He received his Medical degree from the Leyden Medical School in 1938 and a PhD from the University of Groningen in 1946.
- As a medical student, he witnessed a 22-year-old man die from kidney failure and this inspired his research.
- In 1940, he organised the first blood bank in Europe and by 1941 had developed the first artificial kidney machine.
- From 1950–67 he was Professor of clinical investigations at the USA Cleveland Clinic Foundation, and in 1967 he became director of the artificial organs division at the University of Utah, USA.
- Dialysis machines currently keep more than one million patients alive, worldwide.

Function of the kidney

- 1 Complete the sentences using these words: *spine, two, fist, rib, red, water, bones, pressure*.

Humans have _____ kidneys located on either side of the _____.

They are positioned below the _____ cage and are the size of a _____.

Kidneys have many functions, including removing waste, regulating _____ and mineral levels, producing hormones, maintaining healthy _____, assisting in _____ blood cell production (by producing the hormone erythropoietin which aids in the formation of red blood cells), and controlling blood _____.

Definitions

Dialysis machine – Device that cleans and purifies blood when a person's kidneys do not work

Nephron – Functional unit of the kidney

Functions of the kidney

- 2 Referring to figure 1, describe the main functions of the kidney.

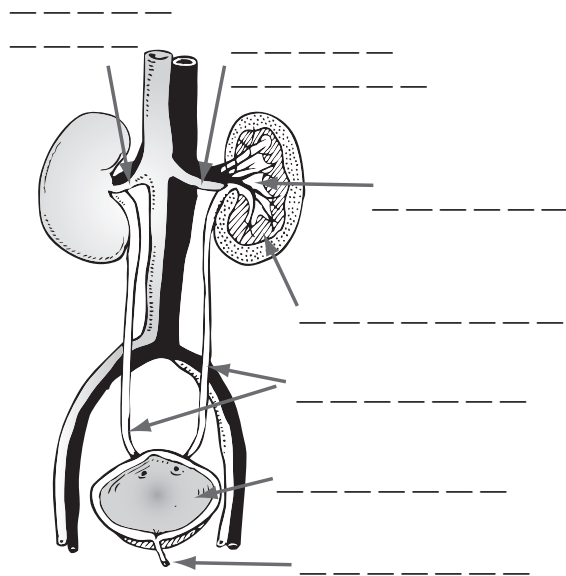


Figure 2 – Structure of the urinary system

Structure of the urinary system

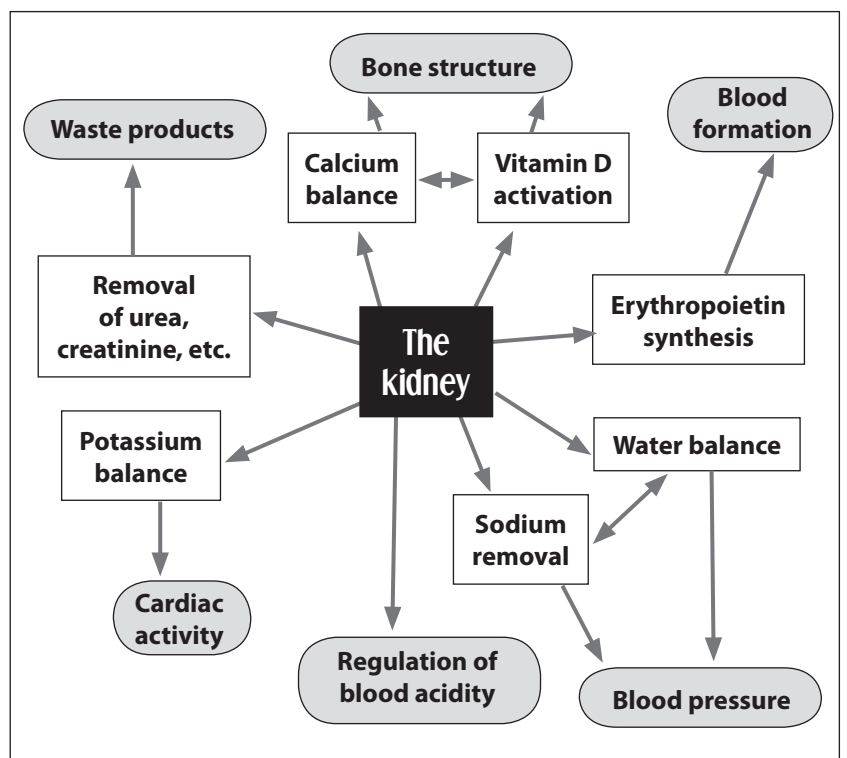
- 3 Label the parts of the urinary system in figure 2 using these words: *pelvis, nephrons, renal artery, urethra, bladder, ureters, renal vein*.

Causes and types of kidney disease

- 4 Complete the sentences by filling in the missing vowels.

Chr_n_c k_dn_y d_s_as_ is c_us_d by m_ny f_ct_rs, _ncl_d_ng d_abet_s, h_g_h bl_d pr_ss_r_, _nf_ct_ _ns, phys_c_l inj_r_y, _ld _g_ _nd h_r_d_t_r_y f_ults. There are also m_ny typ_s _f k_dn_y d_s_s_.

Figure 1 – Functions of the kidney



5 Write each disease beside the correct description in table 1: *polycystic kidney; kidney stone, kidney infections, kidney cysts, kidney cancer.*

Table 1 – Types of kidney disease

Disease	Description
	One or many tumours formed in the kidney
	Pouches of fluid that accumulate in the kidney
	Usually caused by bacteria, also called pyelonephritis
	A solid piece of material forms in the kidney. Can be painful and block urine flow
	A genetic disease where cysts grow in the kidney and lead to kidney failure

Deterioration of kidneys

Once a kidney has been damaged, it will continue to deteriorate.

Table 2 – Effect of medication on the function of a damaged kidney

Time (years)	0	1	1.5	2	3	4	5	6
% kidney function	10	13	25	20	18	13	10	5*

* Commenced dialysis

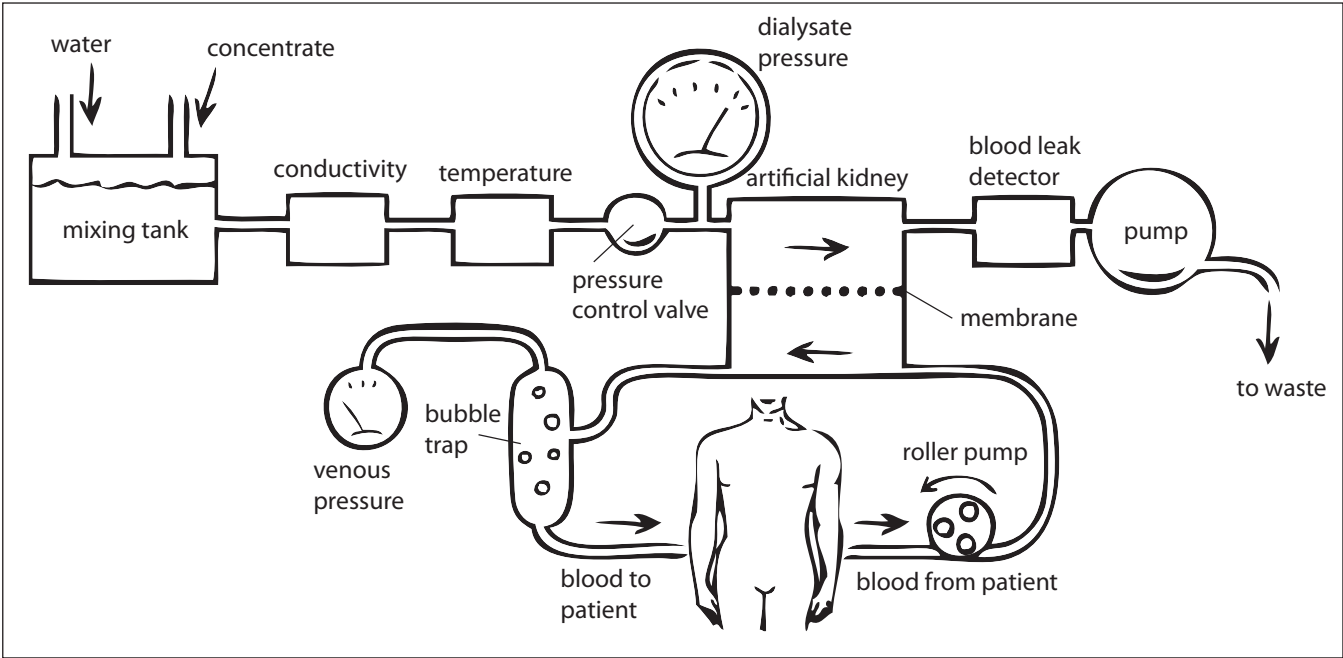
- 6 Graph the data in table 2, and describe the shape of your graph.
- 7 Describe how the medication affected the damaged kidney.
- 8 How many years after initial diagnosis was dialysis initiated?
- 9 Explain why someone may not have a transplant, even though they need one?

Kidney dialysis machine

The artificial kidney has been used for more than 50 years to treat patients with renal failure. Kidney dialysis has four important functions:

- 1 Maintain the correct electrolyte balance within the body
- 2 Maintain pH levels
- 3 Maintain fluid balance
- 4 Clean the blood and remove waste and water
- 10 Study figure 3, which shows the basic structure of a dialysis machine. Outline how the machine purifies blood.

Figure 3 – Dialysis machine



Extension

- * Describe the importance of the kidneys in maintaining the body's processes.
- * Research the history of organ transplants and technological developments in organ repair and replacement.



Edwin Hubble

1889–1953

- Edwin Hubble was born in Mansfield Missouri, USA. In 1910 he completed his Bachelor of Science at the University of Chicago.
- He then completed his Masters at Oxford and his PhD at the University of Chicago in 1917.
- Edwin worked as an attorney in Kentucky and was a major in the army during World War 1. After the war he worked at the Mount Wilson Observatory.
- Edwin discovered that the universe was expanding.
- He classified galaxies by their shape and composition.
- The Hubble Telescope was named in his honour.

Structure of the universe

The universe is the sum of all matter that exists. It consists of super clusters of galaxies, and is believed to be more than 15 billion light years in size.

- 1 Use the information in figure 1 (below right) to draw five diagrams in the spaces provided to show the structure of the universe.

The Milky Way galaxy

- 2 Complete the sentences using these words: *Local, 28 000, Way, stars, spread, shape, spiral, light, solar*.

The Milky _____ is a galaxy and contains our _____ system and 200–400 billion other _____.

Galaxies are not _____ evenly throughout the universe. Instead, they form clumps. Galaxies are classified according to their _____.

The Milky Way is part of 3000 galaxies called the _____ Group.

The Milky Way is a _____ galaxy with a diameter of 100 000 _____ years.

Our solar system is situated within a small spiral arm, and is approximately _____ light years from the centre of the Milky Way.

Structure of the Milky Way

- 3 Use the information in question 2 to draw the structure of the Milky Way

Our solar system

Table 1 (on the next page) lists the characteristics of the planets that make up our solar system. Refer to table 1 to answer the following questions.

- 4 Name the planets that make up our solar system.
- 5 Which planet is closest to the Sun? How close is it?
- 6 State the diameter of the largest planet.
- 7 Which planet has the longest day?

Definitions

Hubble's Law – The further a galaxy is, the faster it is moving

Universe – Sum of all matter that exists

Light year – How far light travels in one year (9 460 000 000 000 km)

Big Bang theory – Belief that the universe was created by a large explosion

Figure 1 – Structure of the universe

1 In our solar system, the Earth revolves around the Sun.	2 Our solar system is part of the Milky Way galaxy.
3 Our galaxy, the Milky Way, is part of a group of galaxies called the Local Group.	4 The Local Group of galaxies make up a bigger cluster of galaxies called the Virgo Cluster.
5 Millions of galaxies cluster together to form the universe.	

- 8 Which planet has the longest year?
- 9 What patterns can you see in the length of one year for the planets?
- 10 Which ‘planet’ was deleted from the list in 2006? Why?
- 11 Draw and label a diagram of the solar system.

Table 1 – The planets of our solar system

Planet	Distance from Sun (million km)	Diameter (thousand km)	Length of day (Earth hours or days)	Length of year (Earth days or years)	No. of moons
Mercury	58	4 880	58.6 days	88 days	0
Venus	108	12 104	243 days	224.7 days	0
Earth	149.6	12 756	24 hours	365.3 days	1
Mars	228	6 787	25 hours	687 days	2
Jupiter	778	143 000	10 hours	11.9 years	16
Saturn	1 427	120 000	10 hours	29.5 years	20+
Uranus	2 875	51 100	23 hours	84 years	16+
Neptune	4 497	49 500	16 hours	165 years	8

Hubble’s Law and universal expansion

Edwin Hubble plotted the velocity at which the galaxies were moving away from the Earth, relative to their distance. He found that the further away the galaxies were, the faster they were travelling.

This discovery helped astronomers calculate the point from which expansion commenced and provided more evidence to support the Big Bang theory.

- 12 Graph the data in table 2 to show Hubble’s Law.
- 13 Describe the shape of the graph and what it shows.

Table 2 – Determining Hubble’s Law

Relative velocity of galaxy (km/s)	Distance of galaxy (mvc)
0	0
10 000	200
20 000	400
30 000	600
40 000	800
50 000	1000
60 000	1200

The Hubble space telescope

The Hubble telescope was named after Edwin Hubble. It was launched on 24 April 1990 from a space shuttle, and has taken more than 250 000 images of the universe. The telescope is 13.2 metres in length, weighs 11 110 kilograms and is 4.2 metres in diameter – which is similar to the size of a large school bus. This huge telescope fitted inside the cargo bay of the space shuttle.

The telescope uses solar energy at 2800 watts. It has two mirrors, the Primary mirror and the Secondary mirror. The Primary mirror has a diameter of 2.4 metres and weighs 828 kilograms. The Secondary mirror has a diameter of 0.3 metres and a weight of 12.3 kilograms.

The Hubble telescope is in orbit, 575 kilometres above Earth’s surface. It completes one orbit every 97 minutes, and will remain in orbit for up to 20 years. The cost of the Hubble telescope project was \$1.5 billion.

- 14 Explain how the telescope was launched into space.
- 15 What are the advantages of viewing the universe from space?
- 16 State the size of the primary mirror of the telescope. How does its height and weight compare to your height and weight?
- 17 The Hubble telescope cost more to build than every major observatory on Earth, combined. How can this cost be justified?

Future telescopes

James Webb telescope

The James Webb space telescope is scheduled to be launched in 2010. A space craft will transport the telescope to a distance of 1.5 million kilometres from the Earth, from where the telescope will orbit the Sun. The telescope will have a mirror with 36 hexagonal segments and will be six times the size of the mirror in the Hubble telescope.

- 18 Explain why a larger mirror increases the effectiveness of a space telescope.

Extension

- * Discusss the reasons why we want to probe and explore our universe.
- * *Space research is essential for understanding why we exist.* Discuss this statement.
- * List arguments for and against this statement: *It is essential that we continue to spend billions of dollars on space research and development.*



Percy Spencer

1894–1970

- Percy was born in Howland in Maine, USA. He was orphaned at a young age and raised by his aunt and uncle. At the age of twelve he worked at a mill and subsequently joined the Navy.
- Percy later worked for the Raytheon Company which built radar devices for the Allied Forces during World War 2.
- While standing in front of a magnetron (the power tube that drives a radar set), he found that a chocolate melted in his pocket. From this observation Percy developed the first microwave oven in 1945.
- More than 200 million microwave ovens are now in use around the world.

Microwaves

Microwaves form part of the electromagnetic spectrum.

- 1 Refer to figure 1 to answer these questions.
 - a What is the wavelength of microwaves?
 - b Can microwaves be observed with the naked eye?
 - c List the main types of electromagnetic radiation.
 - d List two other forms of technology that use microwaves.

Microwave ovens

Microwave ovens contain a magnetron, which converts electricity into microwaves.

The microwaves pass into an enclosed metal cavity and are absorbed by the food or drink.

Water molecules within the food vibrate and molecular friction produces heat, which cooks the food.

- 2 Label the microwave oven in figure 2 using these words: *rotating paddle, wave guide, magnetron, filter, transformer, power, microwaves.*
- 3 List the advantages of the microwave oven.
- 4 Explain why metal containers cannot be used in microwave ovens.
- 5 Outline the reasons why food should not be cooked in regular plastic containers.
- 6 Describe what would happen if you cooked a whole egg in a microwave oven. (Describe it, do not try it!)

Other uses of microwaves

- 7 Outline the uses of microwaves in each of the applications listed below.
 - a Cable television
 - b Mobile phones
 - c Radar
 - d Air and sea navigation
 - e Industrial heating
 - f Therapeutic diathermy treatment
 - g Military, e.g. missile guidance systems
 - h Study of the ozone layer
 - i Masers – similar to lasers, but work at microwave frequencies
 - j Remote sensing of the environment, e.g. vegetation density and groundwater levels

Figure 1 – Electromagnetic spectrum

Produced in	Frequency (Hertz)	Type	Wavelength (m)
Atomic bomb	10^{20}	Gamma rays	10^{-12}
X-ray tube	10^{18}	X-rays	10^{-10}
Light globe	10^{16}	Ultraviolet	10^{-8}
		Visible light	10^{-6}
		Infra-red	10^{-4}
Radiator	10^{14}		10^{-2}
Sun	10^{12}		10^0
	10^{10}	Microwaves	
TV transmitter	10^8		10^2
	10^6	Radiowaves	
Radio	10^4		10^4

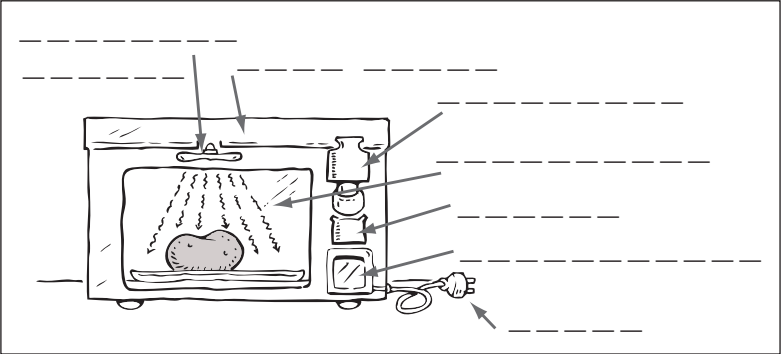
Definitions

Microwave oven – Device which uses microwaves to rapidly heat food

Magnetron – Device that converts electricity into microwaves

Microwave – Form of electromagnetic radiation

Figure 2 – Structure of a microwave oven



Microwaves and mobile phones

- 8 Answer the following questions using what you already know and the information provided in table 1.
- a Outline the main uses of mobile phones.
 - b Explain why mobile phones are so popular.
 - c Suggest reasons why many parents buy mobile phones for their children.
 - d Why is the use of mobile phones banned while driving?
 - e Outline the disadvantages of mobile phones.
 - f Discuss the antisocial behaviour that has developed because of the use of mobile phones.
 - g Explain why people do not want mobile phone relay stations near their homes.

Table 1 – Mobile phone facts

Number of mobile phones worldwide	More than 550 million
Fastest text message	233 character message in 100 seconds
Reaction rate of drivers using a mobile phone	30% slower
Reason for mobile ownership	30% for emergencies
Cancers	Mobile phone users are 2.5 times more likely to develop cancers in areas of the brain adjacent to their ear, but research has not been conclusive
Positioning systems	Mobile phones can be used to locate a person within a few metres, anywhere in the world

Microwaves and remote sensing

- 9 Complete the following information using these words: *satellites, remote, bushfires, weather, based, thermal, atmosphere, water, land, Antarctica*.
- _____ -sensing technology provides a bird’s eye view of the planet, using data from Earth-orbiting _____, aircraft and ground-_____ sensors.
- Microwave radiation induced by _____ heating is emitted from land, sea and the atmosphere.
- Passive microwave detectors have been used to provide information about the Earth’s surface, oceans and _____.
- Microwave remote sensing has been used to study the locations of _____ in remote locations, predict the _____, map _____ use, determine the location of underground _____ and minerals and measure ice conditions in _____.

Microwaves and radar

The word *radar* is an abbreviation for ‘radio detection and ranging’ and is the use of electromagnetic waves such as microwaves to detect objects.

10 Explain how microwaves are used to detect the speed of a vehicle. Include diagrams in your answer.

11 What do you think the potential military applications of radar are?

True or false?

- 12 Answer true or false to the following statements and explain your reasoning.
- a Microwaves cannot be made from electricity.
 - b Microwaves have a lower frequency than X-rays.
 - c Microwaves are a form of electromagnetic radiation.
 - d Microwaves are not produced by the Sun.
 - e Microwaves can affect a person’s health.
 - f Microwaves are a form of energy.

Extension

- * Discuss the benefits and problems of microwave technology.
- * List arguments for and against the statement: *Mobile phones should be banned until the possibility of a link with cancer has been fully researched.*
- * How would your life be affected if mobile phones did not exist?



James Watson

1928–

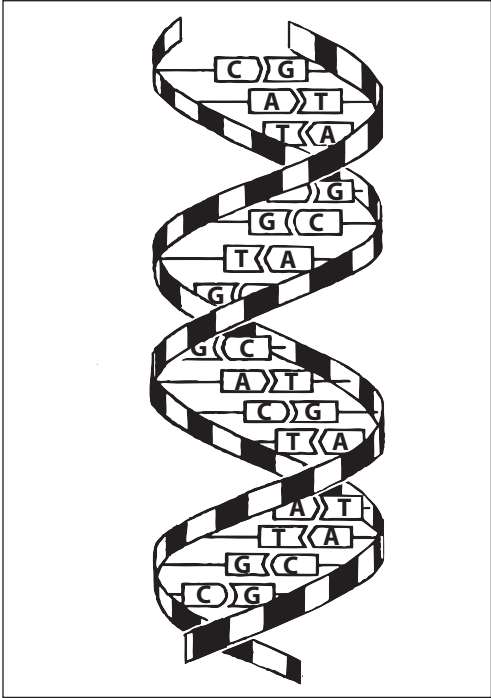
- James Watson was born in Chicago, Illinois USA. When he was 15, he studied Natural Science and Zoology at the University of Chicago, then completed his PhD at the University of Indiana.
- From 1951–53, he worked with Francis Crick at Cambridge University, trying to identify the structure of DNA.
- In 1962, Watson, Crick and Wilkins won the Nobel Prize for discovering that DNA was a double-stranded helix.
- In 1988, James became associate director of the Human Genome Project. The aim was to determine the location and nature of the genes that make up human DNA.
- James established the ethical, legal and social issues program because of his concern for scientific ethics and public misunderstanding of scientific issues.

Structure of DNA

DNA consists of a double helix. The uprights of the ladder consist of alternating sugar and phosphate groups. The rungs consist of paired nitrogen bases.

- 1 Imagine you need to explain to another student the structure of DNA. Colour and annotate figure 1 so that the student can understand this concept.

Figure 1 – Structure of DNA



Chromosomes

Table 1 – Number of chromosomes of different organisms

Organism	No. of chromosomes
Human	46
Chimpanzee	48
Cattle	60
Cat	38
Fruit fly	8
Cabbage	18
Garden pea	14
Goldfish	94

- 2 List the organisms in table 1 in order from largest number of chromosomes to smallest.
- 3 Explain why Watson conducted his first genetic studies on peas and fruit flies.
- 4 Is there a correlation between the number of chromosomes and the size and complexity of an organism? Explain your answer.

Down Syndrome

Table 2 shows the link between the age of a mother and the incidence of Down Syndrome in her child.

- 5 Graph this data.
- 6 Describe the shape of the graph. What conclusions can be drawn from the data?
- 7 What environmental factors could damage a mother's genes?

Definitions

Chromosomes – Strands that contain the genetic material DNA

DNA – Deoxyribonucleic acid, a double helix containing genes and a blueprint of our characteristics

Genes – Sections of DNA

Gender – Male or female

Genetic engineering – To manipulate, modify and transfer the genes of an organism

Table 2 – Effect of age of mother on the frequency of Down Syndrome

Age of mother (years)	Frequency of Down Syndrome per 10 000 births
<20	25
20	25
25	30
30	50
35	75
40	120
45	300

Source: American Family Physician. August 15, 2000

Genetic engineering

- 8 Complete the following information using these words: *engineering, fingerprinting, insulin, duplication, DNA, organ, coding, embryo, paternity, clotting, modified, diseases, replacing.*

Genetic _____ is the ability to manipulate, modify and transfer genes or segments of _____.

Cloning refers to the _____ of an organism, such as Dolly the sheep.

Embryonic stem cell technology refers to the removal of cells from an _____, and the culture of

these cells, which could potentially eliminate the need for _____ transplants.

DNA _____ is the use of electrophoresis to produce a series of bands that represents an individual's DNA. It has been used for identification in criminal investigations and _____ tests.

Gene splicing is the modification and control of genes. This technique has been used to make medicines such as _____, growth hormones and blood _____ factors. To make insulin, the gene _____ for the protein is inserted into a bacterial plasmid, which then multiplies and produces insulin. Gene splicing has also been used to produce genetically _____ foods.

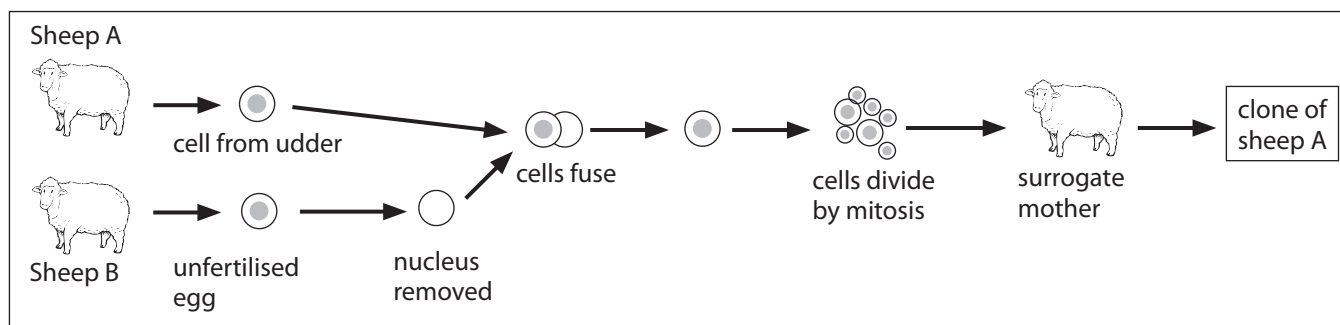
Gene therapy refers to the process of _____ defective human genes with healthy genes to eliminate _____ and other abnormalities.

9 Outline the use of genetic engineering to increase plant production.

10 What moral and/or ethical issues are raised by genetic engineering technology?

Cloning Dolly

Figure 2 – Cloning Dolly



11 Referring to figure 2, outline the major steps involved in the cloning of Dolly the sheep.

Key terms

12 Write each term beside the correct meaning: *gene, chromosome, DNA, genetics, gene mapping, cloning, genotype, phenotype, dominant genes, transgenic, recessive genes, genetic engineering, DNA fingerprinting, human genome project, gene therapy, genetically modified food, biotechnology.*

- a Manipulation of the genetic material of an organism
- b Deoxyribonucleic acid
- c Portion of DNA containing coded information
- d Strands containing DNA
- e Study of heredity and variation in living organisms
- f Making identical copies of an organism
- g Outward appearance of an organism
- h Genes that are not expressed or seen
- i Genes that are expressed and shown with a capital letter

- j Gene from one organism placed in another
- k Used to identify a person, based on a sample of their DNA
- l Genetic make-up of an organism
- m Identifying the location and base pairs of human DNA
- n Replacing an abnormal gene with a normal gene
- o Using organisms to make products, e.g. yoghurt and wine
- p Changing the genetic make-up of food
- q Determining the positions of genes on DNA

Extension

Examine the moral, ethical or legal considerations of one of these genetic engineering projects:

- * cloning humans
- * the human genome project
- * patenting animals
- * genetically modifying food
- * genetically modified crops
- * growing organs such as ears on mice for human transplants
- * DNA fingerprinting
- * gene therapy



Henrietta Leavitt

1868–1921

- Henrietta was born in Cambridge Massachusetts, USA. During her youth she suffered serious illness and was severely deaf.
- In 1895, she did voluntary work at Harvard College Observatory. After seven years, she was given a permanent position and paid 30 cents per hour.
- Henrietta discovered more than 2400 variable stars, half of the total known in her day.
- In 1912, she published a chart of 25 Cepheid stars, indicating their periods and brightness levels. This allowed astronomers to determine distances in space.

Stars

There are hundreds of billions of stars in the Milky Way. Stars consist of 75% hydrogen, 22% helium and traces of oxygen, neon, carbon and nitrogen.

The nuclear fusion of hydrogen within the star provides the fuel for stars to release heat, light and energy. Stars are formed in nebulae, and most stars that we see are 1–10 billion years old.

- 1 Define the word *star*.
- 2 What is the energy source of a star?
- 3 Explain where stars are made.

Importance of stars

- 4 Complete table 1 by providing an example of the importance of stars to each aspect of society.

Definitions

- Cepheid stars* – Stars which change brightness levels in a regular pattern
- Nebulae* – Clouds of dust and gas where stars are formed
- Star* – Mass of hot gases that emits energy waves
- Magnitude* – Degree of brightness of a star

Stages in the life of a star

A star changes its shape and appearance throughout its life. The mass of a star is the principal determinant in its evolution and fate.

When a star has exhausted its hydrogen supply, it burns helium and heavier elements which causes the star to expand and transform into a red giant. When the red giant loses enough fuel, its surface gravity causes it to collapse.

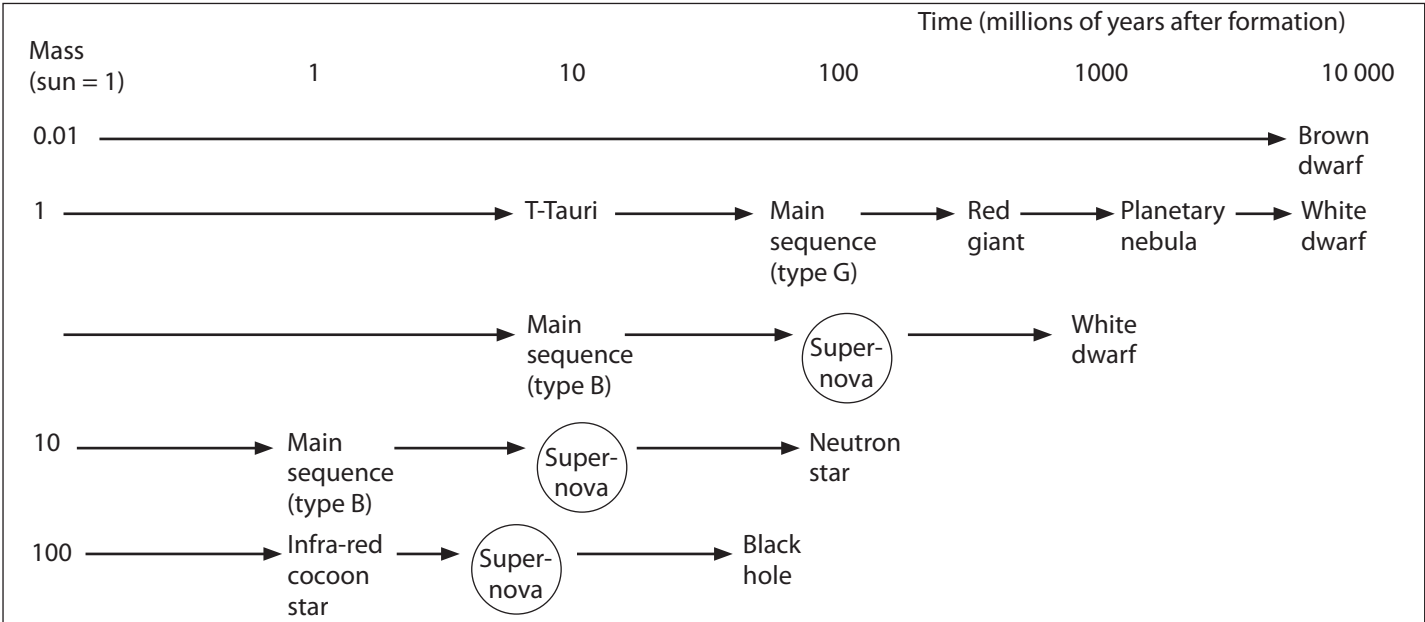
As pressure and temperature increase, the star can explode and become a supernova.

- 5 Referring to the information above and figure 1, discuss the life cycle of a star, based on its size.

Table 1 – Importance of stars to society

Aspect of society	Example of importance
Religious practices	
Astrology	
Navigation	
Calendars	
Children’s myths and stories	

Figure 1 – Life cycle of a star based on its size



Constellations

Prominent stars create patterns. The Babylonians named these constellations after elements of myths, legends and nature.

- 6 a Name the twelve constellations used in astrology.
- b Identify each constellation on a star map.

Age of a star – Hertsprung-Russell (H-R) diagrams

H-R diagrams show the relationship between the temperature of a star and its luminosity.

- 7 Referring to figure 2, list the characteristics of the following types of stars.
 - a Red supergiant
 - b Blue giant
 - c White dwarf
 - d The Sun

The Sun

The Sun is a star. Its characteristics are shown in table 2.

- 8 What is the Sun made of?
- 9 Explain how the Sun derives its energy.
- 10 Why is the Sun essential for our survival?
- 11 Outline the consequences of the Sun running out of fuel.

Careers in the stars

- 12 On the diagram below, list ten careers associated with space research.



Figure 2 – Characteristics of stars

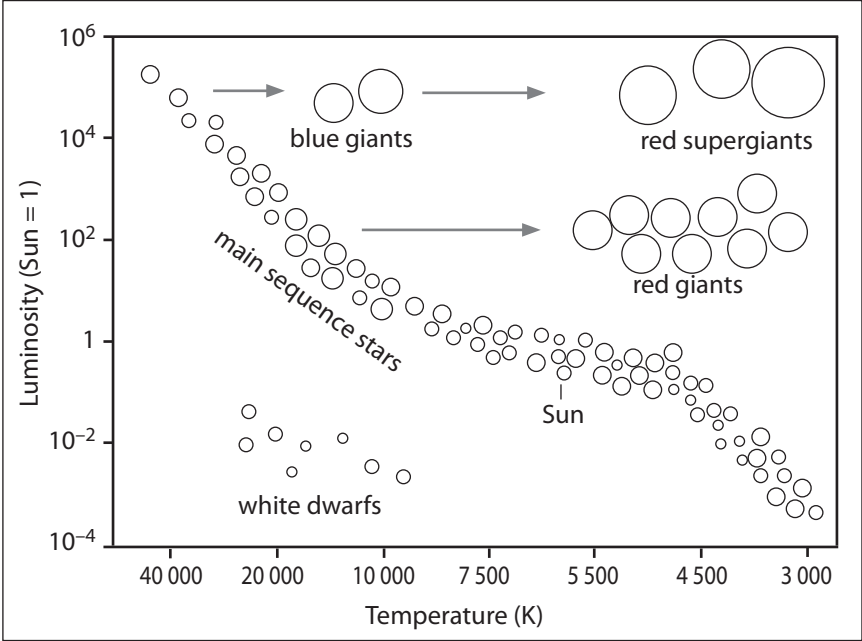


Table 2 – Characteristics of the Sun

Type of star	Main sequence G2
Diameter	1 390 000 km
Temperature	5 800 degrees Kelvin – surface 15 600 000 degrees Kelvin – core
Constituents	70% hydrogen, 28% helium
Period of rotation	Equator 25.4 days; Poles 36 days
Energy production	Nuclear fusion
Sunspots	Cooler regions
Solar wind	A low density stream of charged particles, mostly electrons and protons. Solar winds and solar flares can cause power line surges, radio interference and the colours in the night sky called aurora borealis
Age	4.5 billion years

Extension

- * Apart from the Sun, the nearest star to the Earth is Proxima Centauri which is 39.9 trillion kilometres or 4.2 light years away. Travelling at the speed of a space shuttle (30 000 km/hour), it would take approximately 150 000 years to get there. Outline the current problems involved in space travel. Discuss what other techniques scientists can use to increase understanding of our universe.
- * Find out what constellations and planets are visible at the moment in your area. Arrange an observation night and try matching a constellation map to the real thing.



Bernard Gordon

1928–

- Bernard Gordon was born in Western Massachusetts, USA. He served as a naval officer then obtained a bachelor degree and masters in electrical engineering from the Massachusetts Institute of Technology.
- In 1953, he co-founded the company EPSCO and developed Foetal monitors, Navigation and Traffic Control Systems and the Doppler radar.
- In 1964, Bernard founded Gordon Engineering and invented the CT scanner. Later, he researched Ultrasonography and Digital Imaging.
- In 1992, he founded the Gordon Institute which trains gifted engineers to become leaders in the field.

CT (CAT) scans

- 1 Complete the sentences using these words: *heart, X-rays, CT, technology, rotates, images, osteoporosis, muscle*.

_____ scanners (computerised axial tomography) use _____ and computer _____ to produce slices or 3-D images of the body. They use less X-rays than normal procedures.

The machine consists of a long tube with a narrow bed in the middle. An X-ray gun _____ around the patient and produces 360° _____ of the patient.

Scans show detailed images of organs, bone, _____ and fat. CT scans can identify cancers, blood clots and skull fractures, and determine bone density for diagnosing _____.

Electron beam CTs take pictures in a tenth of a second and show moving organs such as the _____ without blurring.

- 2 Label the structure of the CT scanner in figure 1 using these words: *image, computer, X-ray detector, X-ray tube, patient, X-ray beam, CT scanner*.

Definitions

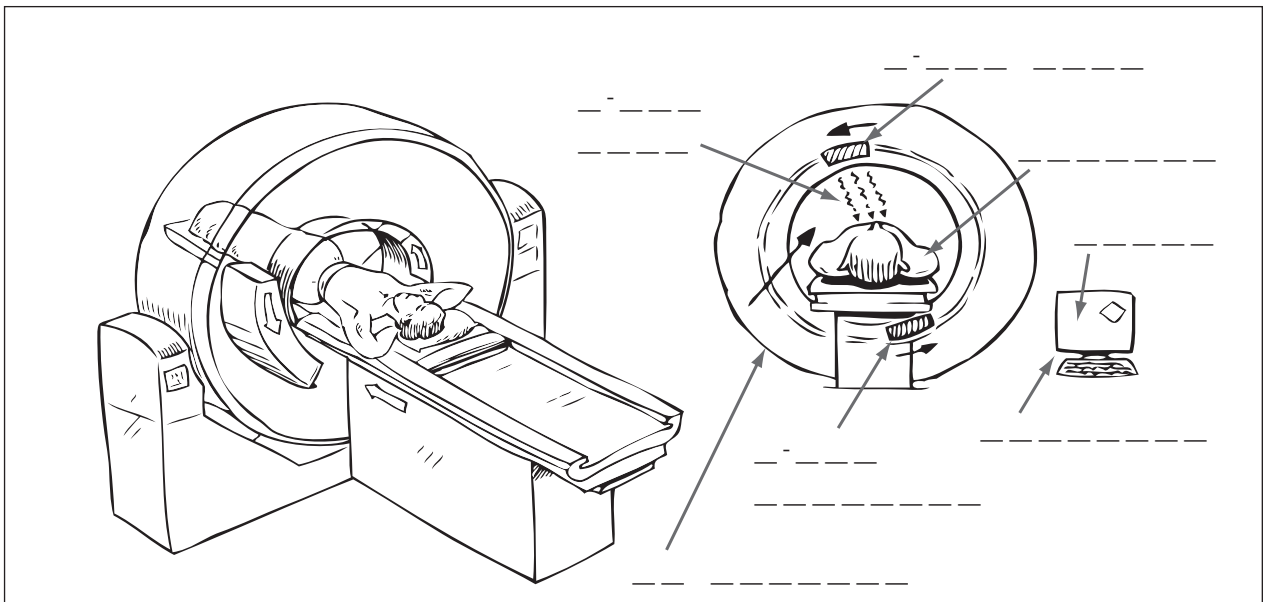
CT (CAT) – Computer axial tomography, a device that uses X-rays to create 3-D images of the body

Doppler effect – Change in the waves in front of and behind a moving object

Foetal monitor – Device used to monitor the heart rate of a foetus

Radar – Detects the location and speed of an object by the reflection of radio or microwaves

Figure 1 – How a CT scanner works

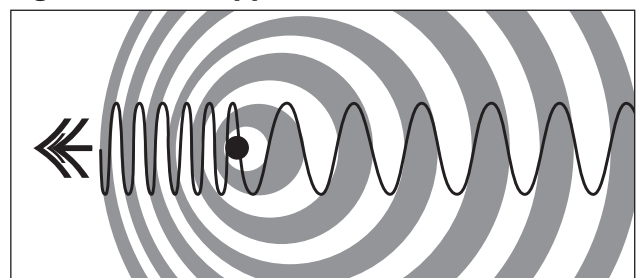


The Doppler principle

The Doppler principle applies to sound, light and radio waves. When a wave rebounds from an object, it changes frequency and can indicate the object's speed and location.

If an object is moving towards the observer, the return frequency will be higher than the original frequency. If the object is moving away, the return frequency will be lower. Figure 2 illustrates the Doppler effect.

Figure 2 – The Doppler effect



- 3 Label the regions of high and low frequency, and the direction of travel, on figure 2.

Doppler radar

The word *radar* is an abbreviation for ‘radio detection and ranging’. Radar is the use of radio or other electromagnetic waves to detect objects.

- 4 Outline in table 1 how each organisation or area might use the Doppler effect and radar.

Table 1 – Uses of Doppler radar

Organisation / Area	Use
Military	
Meteorologists	
Sporting organisations	
Police	
Air traffic control	
Astronomy	

Radar and the weather

Meteorologists use radar to identify weather fronts and storms. Large weather fronts reflect more microwaves and appear as a huge mass on the screen.

- 5 Who would benefit from accurate weather forecasting information?
- 6 Discuss the benefits of early warning systems for hurricanes and other destructive weather conditions.

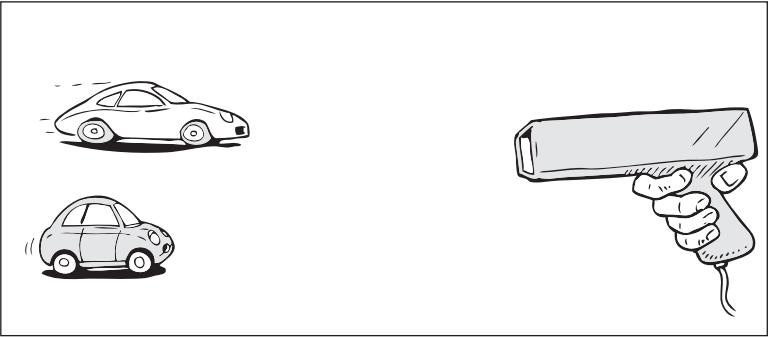
Police and the radar gun

A radar gun contains a radio transmitter and receiver.

Radio waves move at a constant speed. The device can determine location and speed, based on the time it takes for the waves to be reflected from an object.

- 7 Figure 3 shows two cars moving towards a police radar gun. Draw the waves produced by each car.

Figure 3 – Cars moving towards radar



Definitions of key terms

- 8 Write each term beside the correct definition: *Doppler effect, foetal monitor, radar, meteorology, CT scanner, transmitter, military.*
- a** Emits radio waves.

b Shape of waves in front of and behind a moving object.

c Device used to measure location, distance and speed.

d Uses X-rays to show 3-D views of the body.

e Device used to monitor foetal heart rate and stress level

f The study of weather and the atmosphere

g Army, Air Force and Navy

True or false?

- 9 Answer true or false to each statement, and explain your reasoning.
- a** Radar and sonar can direct artillery fire.

b Radar will not work in bad weather.

c Red cars get more speeding tickets.

d Radar and sonar can track aircraft and missiles.

e Radar and sonar can determine the positions of submarines.

f Radar and sonar cannot aid navigation.

g Commercial planes use radar for monitoring and landing.

h Radar cannot be used to study the solar system.

Extension

- * Research sonar, its history and applications.
- * Discuss the use of radar and sonar for commercial and military operations



Gertrude Elion

1918–1999

- Gertrude Elion was born in New York, USA. When she was 15, she entered Hunter College, majoring in science and chemistry. In 1941 she obtained her Masters degree from New York University.
- During World War 2, Gertrude did analytical work for a food company.
- From 1944–83, she worked at Burroughs-Wellcome and developed drug treatments for leukaemia, organ transplants and diseases such as lupus, hepatitis, arthritis and gout, which have saved the lives of millions of people.

Cancer

Cancer refers to the uncontrolled proliferation of cells. It is one of the most common causes of death in the world.

- 1 Name five common types of cancer.

Causes of cancer

- 2 Graph 1 shows the death rates for selected cancers in Australia. What was the major cause of death from cancer?
- 3 What was the major cause of death in males?
- 4 What was the major cause of death in females?
- 5 What factors could increase the likelihood of developing lung cancer?
- 6 Graph 2 shows the incidence of selected cancers in Australia in 2001. Which two forms of cancer had the greatest incidence?
- 7 How could the incidence of melanoma be reduced?

Factors that promote cancer

- 8 Complete table 1 (below right) by explaining how each factor can promote cancer.
- 9 Explain how education can reduce the incidence of cancer, worldwide.

Leukaemia

- 10 Complete the sentences using these words: marrow, Greek, red, circulatory, cells, platelets, leukaemia, genetic, females, death.

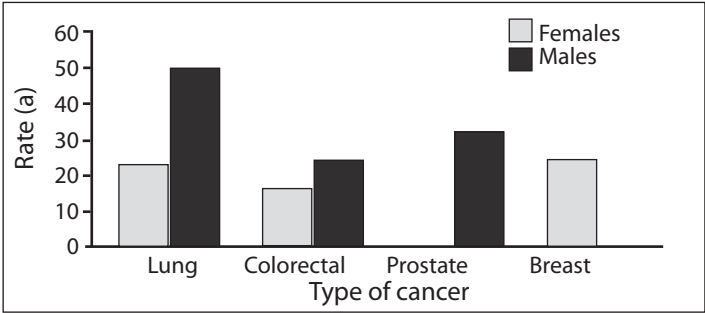
Leukaemia means ‘white blood’ in _____, and refers to the cancer of the _____ system.

This cancer starts in the organs that make blood, namely the bone _____.

An excessive number of immature and abnormal white blood _____ are produced. This results in insufficient numbers of normal _____ blood cells, _____ and white blood cells, and _____ can occur.

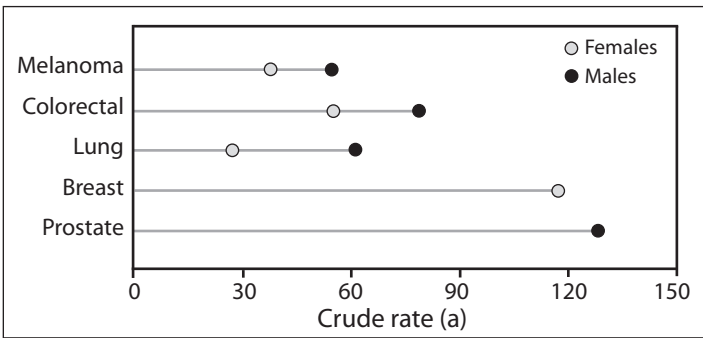
The cause of _____ is unknown. However, it occurs more often in males than in _____. Suspected causes include radiation, chemicals, viruses and _____ factors.

Graph 1 – Death rate for selected cancers, Australia 2004



(a) crude death rate per 100 000
Source: Causes of Death: Australia, 2004 (cat. no. 3303.0)

Graph 2 – Incidence of selected cancers, Australia 2001



(a) crude death rate per 100 000
Source: Cancer in Australia, 2001 (AIHW & AACR, 2004)

Table 1 – Factors that promote cancer

Factor	Reason it promotes cancer
Exposure to the sun	
Diet	
Obesity	
Level of exercise	
Occupation, e.g. working in a chemical industry	
Genetic faults	
Nuclear warfare	
Nuclear reactor accidents	

Symptoms of leukaemia

11 Referring to the list below, draw where the symptoms of leukaemia occur on the person in figure 1.

Symptoms of leukaemia

* bruising

* bleeding

* anaemia

* fever

* chills

* flu-like symptoms

* frequent infections

* weakness

* tiredness

* headaches

* bleeding gums

* pain in joints

* swelling in the abdomen

* swollen lymph nodes, especially in the neck and armpit

* vomiting

* confusion

* loss of muscle control

* seizures

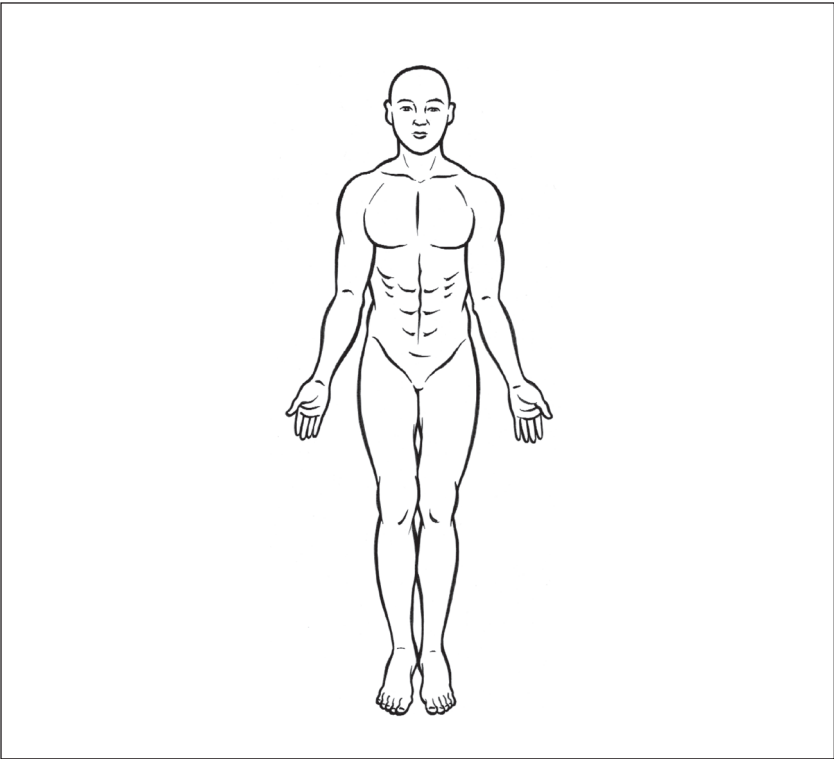
* weight loss

* loss of appetite

* enlarged liver and spleen

* blood in urine or stools

Figure 1 – Symptoms of leukaemia



Treating cancer

- 12 Write each control method beside the correct meaning: *natural remedies, chemotherapy, radiation therapy, bone marrow transplants.*

a Removing diseased marrow and replacing it with healthy marrow

b Use of X-rays or other high energy rays to kill cancer cells

c Use of drugs to kill cancer

d Natural plant and animal products, diet and meditation
- 13 Suggest why doctors are hesitant to recommend natural remedies for the control of cancer.
- 14 Copy and complete table 2 by listing 10 chemicals which have benefited society (e.g. vaccines).

Table 2 – Chemicals that have benefited society

Chemicals/Drugs	How they help	Benefit to society

True or false?

- 15 Mark each statement as true or false. Then correct the false statements so that they are true.

a Leukaemia is a form of cancer.

b Leukaemia is age specific.

c Survival rates from leukaemia have tripled in the last 30 years.

d Leukaemia is a genetically linked disease.

e Leukaemia results in the formation of abnormal red blood cells.

f Genetic diseases such as Down Syndrome make the person more susceptible to leukaemia.

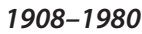
g Dangerous chemicals such as benzene and radiation can increase leukaemia rates.

h Smoking can cause leukaemia.
- 16 List ten careers associated with cancer.

Extension

- * The incidence of cancers is increasing because of toxic chemicals, pollutants and modern technology. Discuss this statement.

* Design a brochure or PowerPoint presentation, promoting healthy lifestyle tips for reducing the chances of developing cancer.



- © Anne Clark and Blake Education 2008

6 What is the half life of carbon-14?

Table 2 – Decay of carbon-14

% carbon-14	Years
100	10
50	5 730
25	11 460
12.5	17 190
6.25	22 920
3.12	28 650

Uses of carbon dating

- 7 List examples of how carbon dating is used in each of these fields.
- a archaeology
 - b geology
 - c geophysics
 - d hydrology and oceanography
 - e antique valuations
 - f proof of identity, e.g. artworks

Definitions

- 8 Write each word beside the correct meaning: *carbon dating, organic compounds, isotopes, carbon cycle, half life.*
- a The conversion of carbon into different forms in nature
 - b Determining the age of a sample by the amount of C-14 present
 - c Amount of time for half of a sample to decay
 - d Chemicals containing carbon
 - e Element containing a varying number of neutrons

Accuracy of carbon dating

The accuracy of carbon dating is based on two main assumptions:

- * The ratio of C-12 to C-14 in the atmosphere has remained constant throughout history.
 - * The rate of decay of carbon has remained constant in the past.
- 9 For each factor listed below, decide whether it would increase or decrease the accuracy of carbon dating. Include reasons for your answer.
- a Radio carbon is forming 28–37% faster than it is decaying (ratio is higher today and hasn’t reached equilibrium)
 - b The industrial revolution released large amounts of carbon dioxide
 - c Burning of forests has released large amounts of carbon dioxide
 - d Volcanic eruptions have released carbon dioxide
 - e Steady decay in Earth’s magnetic field in the last 140 years has increased radiocarbon production

True or false?

- 10 Answer true or false to the following statements, and explain your reasoning.
- a Carbon dating can be used to date non-living things.
 - b C-12 is a radioactive isotope.
 - c Inaccuracies are possible with carbon dating.
 - d The half life of C-14 is about 6000 years.
 - e The industrial revolution and nuclear testing have disrupted the carbon cycle.
 - f Carbon is neither created nor destroyed, only converted into different forms.

Carbon dating and careers

- 11 List ten careers associated with carbon dating techniques.

Extension

- * Discuss the benefits and limitations of carbon dating to society.
- * Research the use of trees and the age rings in their trunks to verify the accuracy of C-14 dating.



Frank MacFarlane Burnet

1899–1985

- Frank Burnet was born at Traralgon, Australia. He gained a medical degree from the University of Melbourne and a PhD from the University of London.
- At the Walter and Eliza Hall Institute in Melbourne, Frank studied microbes and antibody production. His techniques of growing viruses in fertile eggs are still used today to make vaccines.
- In 1960, he and Peter Brain Medawar were awarded the Nobel Prize for the discovery of acquired immunological tolerance. They determined how cells produce antibodies and their research provided the experimental basis for curing autoimmune diseases and the rejection of organ transplants.

Influenza (the flu)

The word *influenza* is derived from the Latin word *influential*, meaning 'having influence'.

The flu is an infectious disease of birds and mammals caused by viruses of the Orthomyxoviridae family. Huge flu epidemics affected the world in 1918, 1957, 1968 and 1977. The pandemic of 1919 killed more than 50 million people, which at the time was 2.5%–5% of the world's population.

- 1 Define the word *virus*.

Deaths from influenza

Table 1 shows deaths caused by influenza epidemics throughout history.

- 2 Graph the data in table 1. Then describe the shape of your graph.
- 3 Which epidemic caused the greatest number of deaths?
- 4 Describe the potential impact of these epidemics on a society.
- 5 What factors could cause the rapid spread of influenza?

Table 1 – Major influenza epidemics

Pandemic	Date	Number of deaths
Russian flu	1889–90	1 million
Spanish flu	1918–20	50 million
Asian flu	1957–58	1.5 million
Hong Kong flu	1968–69	0.75 to 1 million

Source: World Health Organisation: World Health Assembly 2003 Prevention and control of influenza pandemics and annual epidemics.

Definitions

- Influenza* – Viral flu
Avian influenza – Bird flu
Acquired immunological tolerance – Where foreign antigens become regarded as self by the immune system. Important in curing autoimmune diseases
Virus – A particle consisting of its genetic material and outer protein coat
Replication – Making a copy

Structure of the virus

- 6 Figure 1 shows the structure of the influenza virus. Label the diagram using these words: *genetic material*, *envelope*, *hemagglutinin*, *neuraminidase*.
- 7 Explain why viruses can be classified as living or non-living.

Characteristics of the influenza virus

Influenza is caused by three different viruses. The virus is one thousandth of a millimetre in diameter and consists of genetic material surrounded by a protein coat.

Influenza is spread by airborne droplets. Coughing and sneezing or contact with hands, tissues and articles soiled with discharges will spread the disease.

Influenza has an incubation period of 1–3 days and an infection period of 3–5 days.

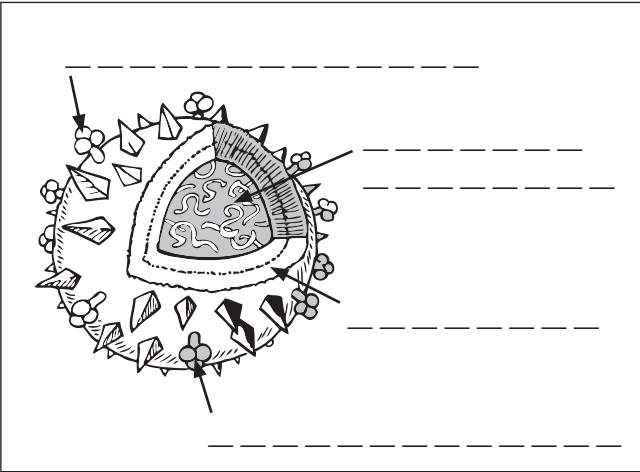
- 8 Explain why the flu can be fatal to the young, elderly or people suffering from chronic illness.
- 9 Why won't previous exposure to the flu prevent subsequent infections?

Symptoms of influenza

10 Referring to the list of symptoms below, label figure 2 to show where the symptoms of influenza occur.

- | | | | |
|---------------------|---|-------------------------------|--|
| * fever | * reddened eyes, skin, mouth, throat and nose | * sore throat and cough | * diarrhoea |
| * headache | | * temperatures of 38.9–40.0°C | * secondary diseases such as pneumonia |
| * extreme coldness | * muscle fatigue | | |
| * nasal congestions | * cough | * nausea and vomiting | |

Figure 1 – Structure of a virus



Control of influenza

There is no cure for influenza. Vaccinations can provide 70% protection, but annual injections are required because the virus mutates.

- 11 Why should a person be vaccinated in autumn, prior to the onset of the flu season?
- 12 How can hygiene and education prevent the spread of influenza?

Avian influenza (bird flu)

- 13 Complete the information using these words: *dollars, avian, humans, cooking, viral, two, bird, mortality, manure, faeces, indirect.*

In 2006, more than ten billion _____ was spent and _____ hundred million birds were killed to try and contain _____ influenza H5N1. Avian influenza is a highly contagious _____ disease that can affect all commercial, domestic and wild _____ species.

The disease has affected many Asian countries and has resulted in 100% _____ in some regions. Migratory birds and waterfowl can carry the virus, but show no signs of the disease. The disease is spread by nasal and eye discharges, contaminated _____ and direct or _____ contact with the birds. The virus can survive for at least 35 days at 4°C in _____ and can be isolated from water where waterfowl are present. The virus can survive in carcasses for up to 23 days if refrigerated, but is destroyed by heating and _____.

_____ in close contact with the birds have been affected by the disease.

- 14 Label figure 3 to show where the symptoms of avian influenza occur.

Figure 2 – Symptoms of influenza

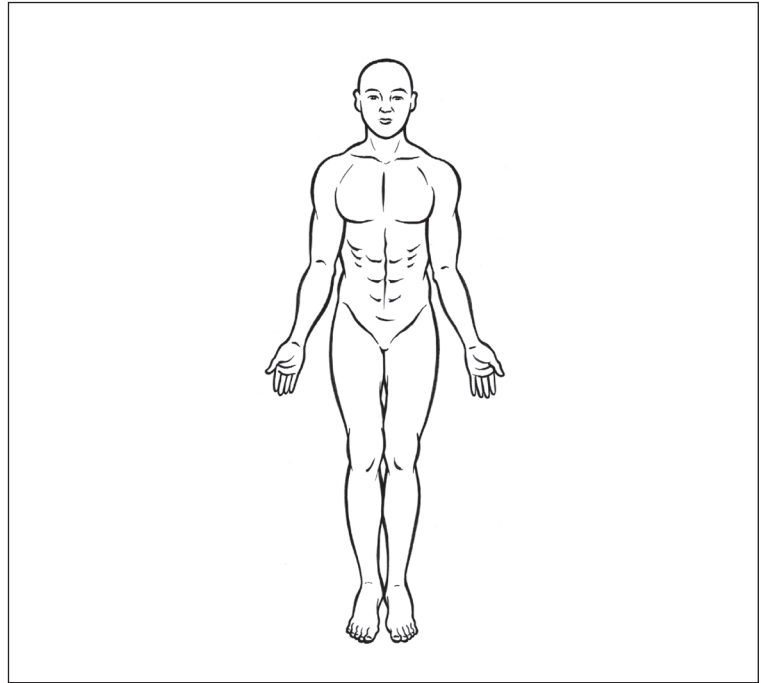
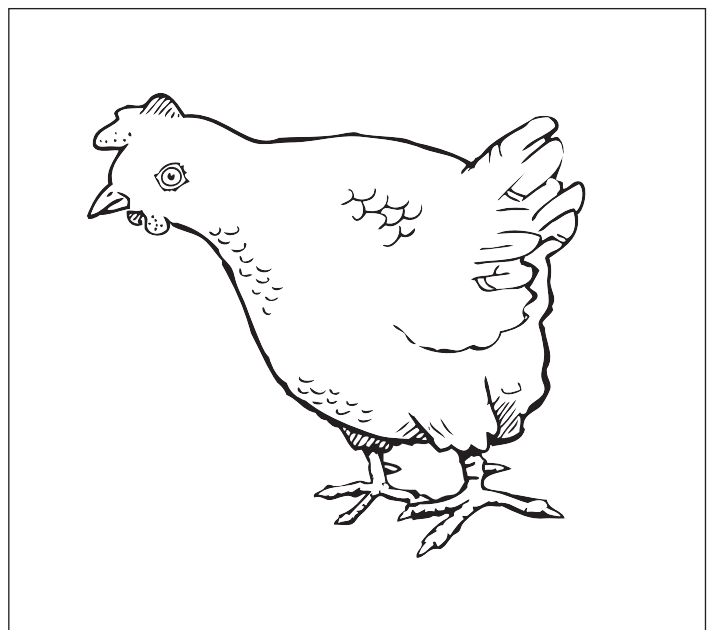


Figure 3 – Symptoms of avian influenza



Symptoms of avian influenza

- | | |
|--|----------------------------------|
| * Depression | * Reddened legs |
| * Decreased egg production | * Difficulty breathing |
| * Soft-shelled eggs | * Diarrhoea |
| * Swollen, bluish-coloured combs and wattles | * Swelling of the face |
| | * 100% mortality within two days |

- 15 Outline how intensive poultry production can increase the likelihood of avian influenza.
- 16 What do you think could be the social and economic effects of this disease on a region?

Extension

- * Discuss the importance of mass vaccination, hygiene, information campaigns, quarantine and culling on the control of avian influenza.

- * Discuss the direct and indirect costs of influenza to society. Include reference to death, lost productivity and cost of research, vaccinations and medical treatment.



Helen Newton Turner

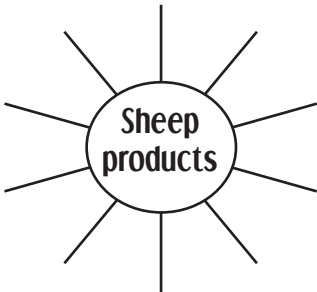
1908–1995

- Helen Turner was born in Sydney, Australia. In 1930 she obtained a Bachelor of Architecture from Sydney University. She studied statistics in England and worked at CSIRO.
- In 1956, she became head of animal breeding and introduced objective measurement into their programs.
- Helen studied the Booroola Merino. She found that a single gene controlled fertility, and obtained lambing percentages of 220%.
- Helen also introduced micro measurement into the wool industry and bred characteristics that consumers wanted.
- Helen worked for the United Nations Food and Agriculture Organisation to assist sheep-breeding programs in developing countries.

Importance of sheep

- 1 Complete figure 1 by listing 10 products derived from sheep.

Figure 1 – Importance of sheep



Definitions

- Heritability* – Measures how strongly a characteristic is passed on from one generation to the next
- Drenching* – Chemical used to kill internal parasites
- Joining* – To mate animals
- Crutching* – To remove the fleece from the rear of a sheep to prevent flystrike
- Marking* – Castration, ear tagging and vaccination
- Objective measurement* – Use of measurement and statistics to select characteristics

Calendar of operations

- 2 Complete table 1 by describing each operation and its purpose.

Table 1 – Calendar of operations on a sheep property

Time period	Operation	Description of operation and purpose of operation
May–June	Ultrasound sheep for pregnancy	
July–August	Lambing	
August–September	Marking	
September–October	Crutching	
November–December	Drenching and weaning	
October–January	Stubble grazing	
December–January	Shearing	
February	Drenching, joining, sale of old sheep	

Wool quality

- 3 Table 2 lists the characteristics that affect wool quality. Write each factor beside the correct meaning: *softness, soundness, fibre diameter, length, colour, lustre, density.*

Products made from wool

- 4 Figure 2 shows the relationship between fibre diameter and end use. State the uses of wool with these fibre diameters:
- a 41 microns
b 17–24 microns
c 28–30 microns
d 34–40 microns
- 5 Write a short paragraph describing the relationship between fibre diameter and its uses. Use examples from figure 2.

Breeding and selection

- 6 Complete the information using these words: *characteristic, next, one, heritability, percentage, breeding, many, gene.*
- _____ is a measure of how strongly a characteristic is passed on from _____ generation to the _____. It is measured as a _____ from 0–100 and indicates what progress can be obtained in a _____ program.
- A _____ governed by one _____ is easier to select and breed for than a characteristic governed by _____ genes.

Table 3 shows the heritability of characteristics for different livestock.

- 7 Outline which characteristic is the easiest to breed for in these animals:
- a sheep b beef cattle
c poultry d pigs
- 8 Describe why each characteristic in table 3 is valuable to a farmer.

Table 2 – Factors affecting wool quality

Factor	Meaning
	Size of the fibre
	38–51 mm is the desirable length for spinning
	Wool that is strong and does not break
	White wool is best for processing and dyeing
	Not harsh and does not provide irritation
	Brightness of the fibre
	How closely the fibres are packed together

Figure 2 – Fibre diameter and end use

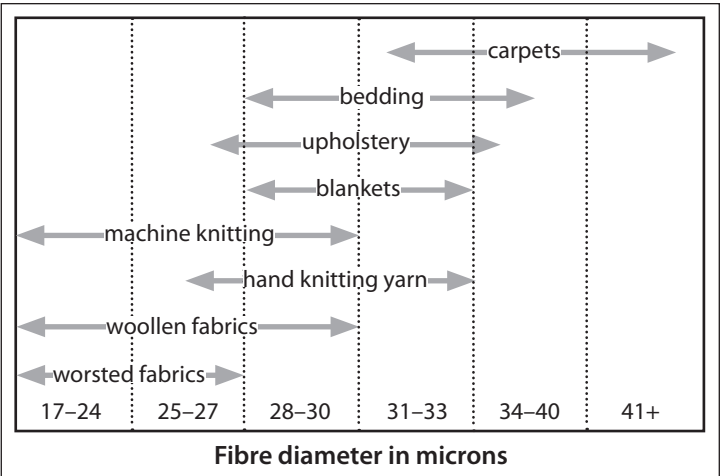


Table 3 – Heritability of characteristics in livestock

Animal	Characteristics	Heritability
Sheep	Fleece weight	35–40
	Fibre diameter	20–45
	Staple length	35
Beef cattle	Efficiency of weight gain	40
	Weaning weight	35
	Post weaning weight gains	35
Poultry	Egg production	20–25
	Egg size	12
	Body weight	45
Pigs	Litter size	0–24
	Litter weight	30
	Post weaning weight gain	14–58

Source Clark A. (2005) *Senior Australian Agriculture*, Pascal Press, Glebe.

Extension

- * Discuss the benefits and the moral and ethical issues of using genetic engineering to breed desired characteristics in animals.
- * Find out how the issues of animal rights and animal welfare affect Australian farmers.



Graeme Clark

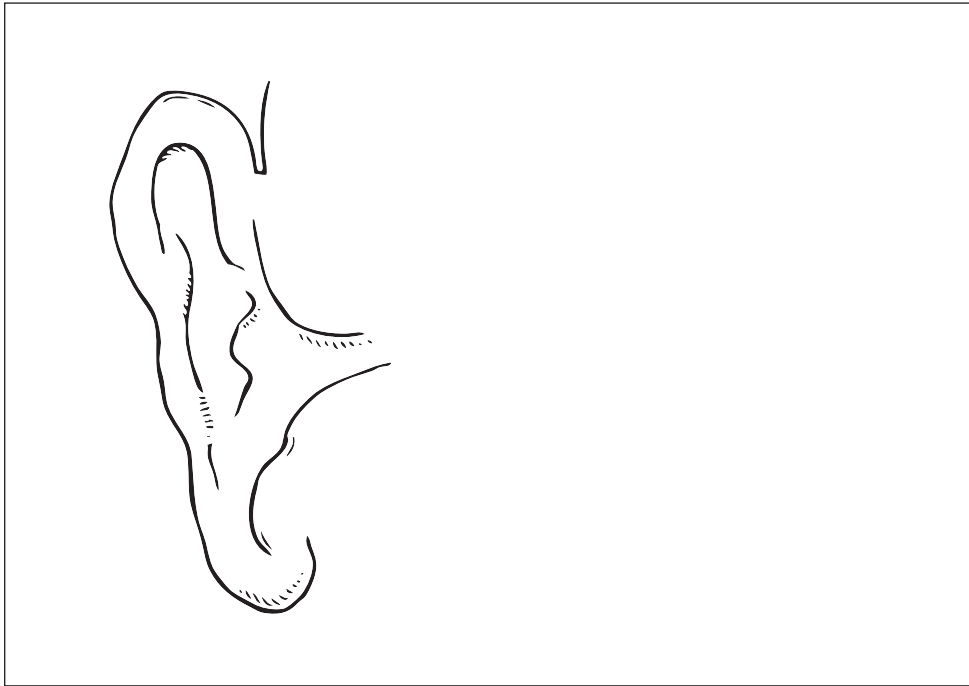
1935–

- Graeme Clark was born at Camden, Australia. He completed his MB, BS, Masters and PhD at the University of Sydney.
- In 1970, Graeme became the Professor of Otolaryngology at the University of Melbourne.
- In 1978, he developed and successfully implemented the first cochlea implant.
- In 1984, he became research director of the Bionic Ear Institute.
- Graeme's invention has resulted in more than 30 000 people in 70 countries receiving the implant.

The ear

- 1 Using the word list, construct an annotated diagram in figure 1 to show the structure of the ear and how sound is heard: *pinna, auditory canal, tympanic membrane, hammer, anvil, stirrup, semicircular canal, cochlea.*

Figure 1 – Structure of the ear



Definitions

Deafness – Complete loss of hearing

Hearing impairment – Loss of hearing from one or both ears

Cochlea implant – Device that allows deaf people to hear

- 2 Explain how sound is detected.

Deafness

- 3 Complete the information using these words: *hearing, globally, hereditary, loud, population, deafness, middle, jaundice, age.*

Hearing impairment is the loss of _____ from one or both ears, whereas _____ refers to a complete loss of hearing. In 2006, the World Health Organisation estimated that 278 million people worldwide had moderate-to-profound hearing loss in both ears. _____, it was estimated that 0.1% of the _____ was deaf.

80% of deaf and hearing impaired people live in low and _____ income countries.

The main causes of deafness include _____ faults, premature birth, insufficient oxygen to the baby during birth, diseases such as rubella, syphilis, _____, meningitis, measles and mumps, chronic ear infections, old _____ and excessively _____ noises.

Genetic disorders

- 4 Figure 2 (next page) shows the inheritance of deafness in a family. Identify the cause of the deafness.
- 5 Define the word *carrier*.
- 6 Explain why one son in figure 2 is deaf.

Sound levels

Loud noises such as gunshots, firecrackers, explosions and rock concerts can damage ears. In air, sound pressure levels above 85 db are harmful and levels above 130 dB cause immediate perforation of the eardrum.

- 7 Referring to table 1, determine which of the following could affect a person’s hearing.
- a jackhammer
- b jet engine
- c low calibre rifle
- 8 a List the activities from table 1 that usually require safety equipment to be used.
- b Comment on the use or non-use of safety equipment at the 90–110 dB range. What are the implications for ear damage of regular exposure to these kinds of activities?

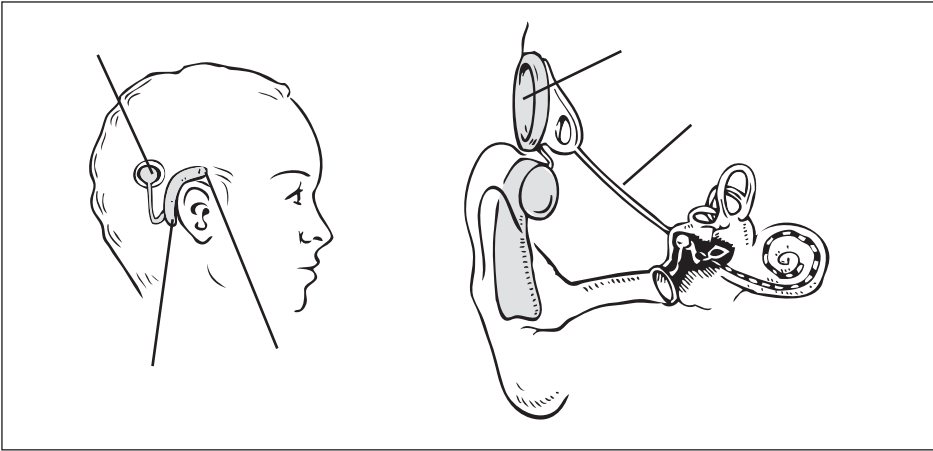
Cochlear implant (bionic ear)

A cochlear implant converts sound vibrations into electrical stimuli. These are detected by the auditory nerve and transmitted to the brain. The cochlear implant consists of a microphone, speech processor, transmitter, receiver and electrodes.

The microphone detects sound and the speech processor converts the sound into electrical signals. The transmitter directs the signals to the receiver, which converts them into electrical impulses. These are relayed to the cochlea which transmits them directly to the brain.

- 9 Label the structure of the cochlea implant in figure 3 using these words: *implanted receiver, electrode system, transmitter, sound processor, microphone*.

Figure 3 – Cochlear implant



- 10 Outline how each factor could limit the use of cochlea implants.
- a Cost of device
- b Surgery
- c Therapy required after implantation
- d Risk of meningitis after operation
- e Not as readily accepted by old patients
- f Children with implants not taught sign language and isolated from deaf children

Figure 2 – Inheritance of deafness in a family

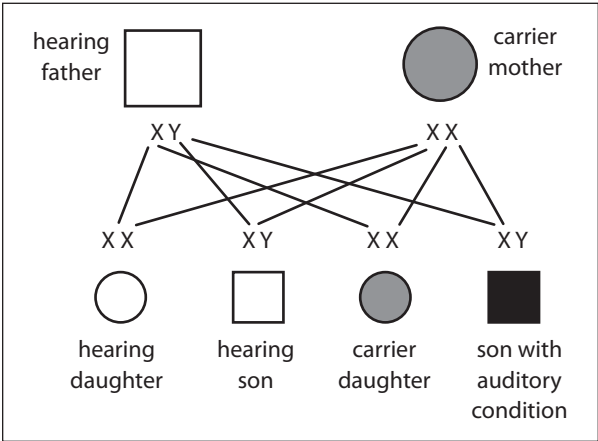


Table 1 – Sound levels of different events

Factor	Decibel range (dB)
Weakest sound heard	0
Human breathing at 3 m	10
Whispering	20
Residential area at night	40
Normal conversation at 2–3 m	60
Busy traffic at 5 m	70
Vacuum cleaner at 1 m	80
Loud factory, heavy truck at 1 m, level at which hearing loss occurs with sustained exposure to the sound	90
Jackhammer at 2 m, inside a nightclub	100
Train horn at 1 m, perforation of eardrums	120
Threshold of pain, emergency siren at 30 m	130
Low calibre rifle being fired at 1 m, the engine of a Formula One car at 1 m	140
Jet engine at 30 m	150
Death of hearing tissue	180
Loudest sound possible	194

Extension

- * Outline the discrimination that deaf people may encounter in education, the workplace and society. How could this issue be addressed?



Ian Frazer

1953–

- Ian Frazer was born in Scotland. He trained at Edinburgh as a Physician and specialised in immunology.
- In 1980 he came to Australia for a three month working holiday then emigrated to Melbourne.
- In 1985 Ian founded the Centre for Immunology and Cancer Research at the University of Queensland. For 20 years he researched the link between papilloma viruses and cancer.
- In 2005 his research group developed a vaccine against the virus.
- The vaccine Gardasil and Cervarix can prevent cervical cancer, which affects 500 000 women per year.

Cervical cancer

- 1 Complete the information using these words: *virus, cervix, uterus, born, teenager, cancer, immune, death, transferred, hundred, lining, rate.*

Cervical _____ is caused by the papilloma _____, and is a tumour or growth that develops in the tissue of the _____.

The cervix is the lowest part of the womb or _____ through which a baby passes when it is _____. Most cancer originates from the surface cells _____ the cervix.

Cervical cancer occurs at any age and can result in _____.

Risk factors include:

- sexual intercourse as a _____
- having Aids or problems with your _____ system
- having many sexual partners, as the virus can be _____ during sex.

Pap testing has reduced the incidence and death _____ of cervical cancer in developed countries. However, the disease still kills several _____ thousand women per year, worldwide.

Definitions

Cervical cancer – Cancer of the reproductive organs caused by the papilloma virus

Vaccine – Dead or diluted form of virus injected into the body to build antibody levels and disease resistance

Papillo mavirus – Microbe that causes cervical cancer

Cervix – Lowest region of the uterus through which a baby passes during birth

Pap test – Smear taken from cervix and checked for abnormal cells

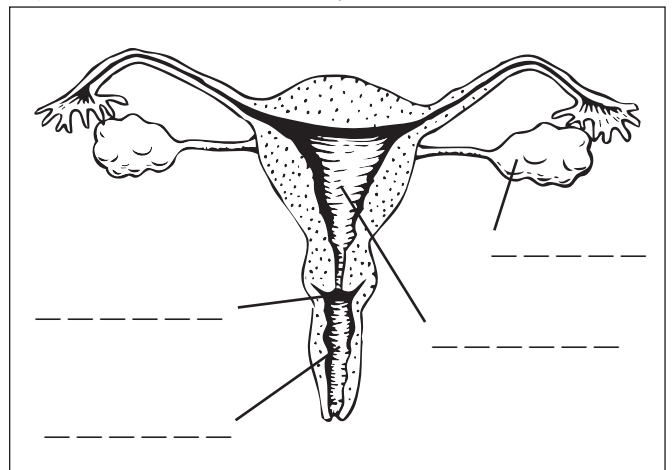
Key terms and meanings

- 2 Write the correct term beside each meaning: *vaccine, HPV, pap smear, cervical cancer, virus, cervix, safe sex.*
- a** Use of preventative methods such as condoms to reduce the spread of papilloma virus
 - b** Particle consisting of DNA and protein coat
 - c** Test used to identify abnormal cells in the uterus
 - d** Region of the uterus through which a baby passes during birth
 - e** Disease of the cervix which can result in death
 - f** Papilloma virus that causes cervical cancer
 - g** Dead or diluted form of disease injected to increase immunity to a disease

Reproductive system of a female

- 3 Label the reproductive system of a female in figure 1 using these words: *ovary, uterus, vagina, cervix.*

Figure 1 – Reproductive system of a female



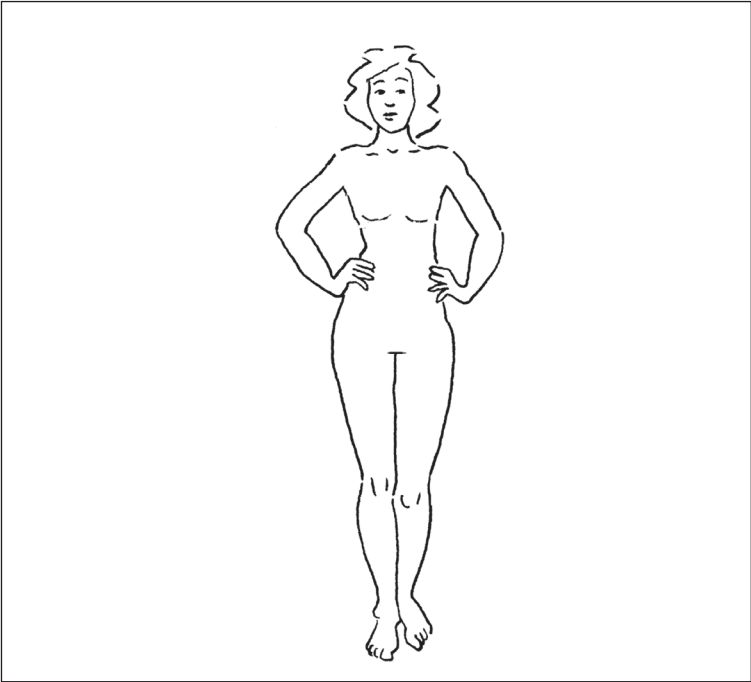
Symptoms of cervical cancer

4 Label figure 2 to show where the symptoms of cervical cancer occur.

Symptoms of cervical cancer

- * Pap test has abnormal cells
- * Bleeding between periods and after menopause
- * Vaginal discharges
- * Lower back pain
- * Difficult urination
- * Possible kidney failure
- * Urine contaminated with blood
- * Swelling of the legs
- * Diarrhoea
- * Fatigue
- * Loss of appetite and weight

Figure 2 – Symptoms of cervical cancer



Age and incidence of cervical cancer

Table 1 shows the effect of age on the prevalence of HPV (papilloma virus) and cervical cancer.

- 5 Graph the data. Then describe the shape of your graph.
- 6 What conclusions can be drawn from this data?

Table 1 – Effect of age on the incidence of HPV and cancer

Age (years)	HPV prevalence %	Cancer incidence per 100 000
15–19	16	0
20–24	25	3
30–34	13	13
35–39	9	15
40–44	8	15
45–49	3	17
50–54	9	16

Source World Health Organisation: Comprehensive cervical cancer control 2006

Incidence and mortality from cervical cancer

Table 2 shows the incidence and mortality rate of cervical cancer in the world

- 7 Present the data in table 2 as a bar graph.
- 8 Which regions had the highest mortality rate from cervical cancer?
- 9 Suggest why the United States has the lowest rates.
- 10 Explain why early detection is vital in preventing cervical cancer.

Table 2 – Incidence and mortality rate of cervical cancer in the world

Region	Incidence per 100 000	Mortality per 100 000
Central America	45	18
South Africa	40	22
South America	30	15
South-east Asia	20	10
Western Europe	12	5
United States	10	5

Source: World Health Organisation - Comprehensive cervical cancer control 2006

The vaccine

The vaccine consists of the outer shell or coat of the virus and stimulates the body’s immune system to produce antibodies that can fight the disease.

Women aged 9–26 years can be vaccinated, and governments are supporting the use of the vaccine.

- 11 Explain why women should continue to have pap smears after they have been vaccinated.

Extension

- * Discuss the strategies that can be used to stop cervical cancer.
- * Explain how the incidence of cervical cancer could be reduced in developing countries. Outline the problems or limitations that would have to be addressed to ensure the program is successful.



Crossword of legendary science

Across

- 1 Discovered that more than one strain of virus caused polio
- 3 Developed DNA fingerprinting
- 7 The feeding relationships in a community
- 8 Electromagnetic waves used to transmit television images
- 10 Invented the first plastic
- 11 Showed a relationship between energy, mass, time and distance
- 15 Along with Francis Crick and Maurice Wilkins, he won the Nobel Prize for discovering the structure of DNA
- 16 Device used to generate energy from the Sun
- 17 Invented the kidney dialysis machine
- 21 An astronomer who discovered 2400 variable stars
- 23 Masses of hot gases that emit energy waves
- 24 Showed that the universe was expanding
- 26 Invented the foetal monitor, Doppler radar and high speed analogue to digital converters used in computers
- 28 Introduced objective measurement practices into animal breeding programs
- 30 The uncontrolled proliferation of cells; one of the most common causes of death in the world
- 32 Form of electromagnetic radiation
- 33 Cancer of the reproductive organs caused by the papilloma virus
- 34 A large mass can warp space and ____.
- 35 The use of measurement and statistics to select characteristics: ____ measurement
- 36 Biodegradable plastics
- 37 An oral vaccine for polio
- 38 An infectious disease affecting birds and mammals that is caused by viruses of the Orthomyxoviridae family
- 39 The detection of location and speed of an object by the reflection of radio or microwaves

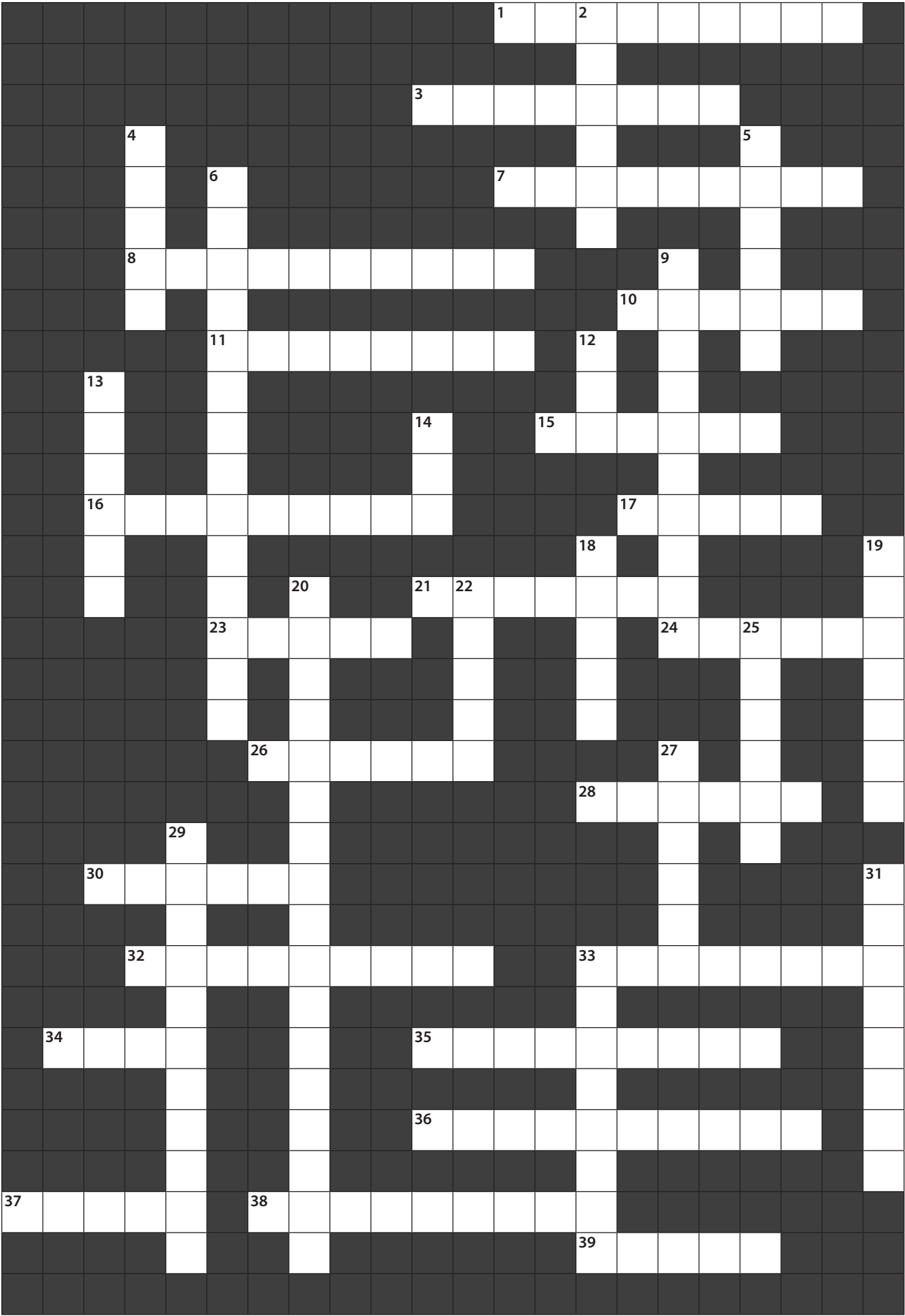
Down

- 2 Author of the book 'Silent Spring', which raised environmental consciousness worldwide
- 4 Developed the cochlear implant
- 5 Physicist and radio engineer who discovered that the Milky Way emitted radio waves
- 6 Device used to treat people with renal failure: ____ machine
- 9 First person to develop and patent the television
- 12 A chemical containing the genetic material of an organism
- 13 A neutron star that spins rapidly and emits pulses of radio waves along its magnetic axis
- 14 Invented the silicon solar cell
- 18 Won the Nobel Prize for developing the radiocarbon dating technique
- 19 Invented the microwave oven
- 20 The use of the isotope carbon-14 to determine the age of materials that contain carbon
- 22 Developed drug treatments for leukaemia, organ transplants and diseases such as lupus, hepatitis, arthritis and gout
- 25 Won the Nobel Prize for pioneering work with microbes and immunology
- 27 Developed a vaccine to prevent cervical cancer
- 29 The manipulation, modification and transfer of the genes of an organism is genetic ____
- 31 The galaxy to which our solar system belongs
- 33 An implant consisting of a microphone, speech processor, transmitter, receiver and electrodes

Word list

BIOPLASTIC	DNA	GORDON	LEAVITT	PARKES	SOLAR CELL
BURNET	EINSTEIN	HUBBLE	LIBBY	PULSAR	SPENCER
CANCER	ELION	INFLUENZA	MACNAMARA	RADAR	STARS
CARSON	ENGINEERING	JANSKY	MICROWAVE	RADIO WAVES	TIME
CERVICAL	FARNSWORTH	JEFFREYS	MILKY WAY	RADIOCARBON	TURNER
CLARK	FOOD CHAIN	KIDNEY DIALYSIS	OBJECTIVE	DATING	WATSON
COCHLEAR	FRAZER	KOLFF	OHL	SABIN	

Crossword of legendary science *cont'd*



Selected answers

Rachel Carson (pages 4–5)

- 1 pesticide, insects, malaria, typhus, odourless, ingestion, muscle, death, resistance, contaminated, bees, chains, cancer
- 2 Carbon, hydrogen, chlorine
- 3 White crystalline substance, tasteless and odourless so difficult to detect.
- 4 1–15 years, longer in some regions
- 5 In fat cells
- 6 °C means degrees Celsius; mg/kg means milligram per kilogram; µg/L means micrograms per litre.
- 7 By air and water particles
- 9 Killed them
- 10 DDT accumulated through each level of the food chain, so the higher the level in the food chain, the higher the level of DDT contamination.
- 12 Disrupts all food chains and webs
- 13 House fly – small mass
- 14 150–750 mg/kg, when ingested
- 15 Eating food contaminated with DDT

Alec Jeffreys (pages 6–7)

- 2 (In table order)
 - can prove if a person is a relative by DNA
 - can identify mother or father by DNA
 - can use DNA from crime to identify criminal
 - can identify suspects or victims by samples of their DNA
 - can identify similar organisms by their DNA
 - unknown soldiers or civilians can be identified
 - defective or faulty genes can be identified and removed
 - can identify scroll fragments that belong together
- 3 a criminal b DNA c Nazi, died d murder e 18 f prison, testing
- 4 All reduce the reliability of the evidence.
- 5 No. More evidence should be used to increase the reliability.
- 6 a Grandfather b Two c Three
- 8 a 50% b No c Six children have been affected: numbers 3,5,7,8,10,11
- 9 It is morally and ethically wrong. However, their priority is profit because they are a business.

Karl Jansky (pages 8–9)

- 3 Sound waves are longitudinal waves and cannot travel through a vacuum. Radio waves are electromagnetic waves and can travel through a vacuum.
- 4 Radio, distances, Milky, pulsars, Bang, temperatures, radiation, magnetic, flares, supernovas
- 6 Radio and television stations cause disturbances and interfere with reception.
- 10 1 second

Russell Ohl (pages 10–11)

- 1 Sun, 10, coal, energy, solar, weather, wave, ancient, metal, current, less, silicon, energy
- 2 Gradually increasing, as technological advances increase efficiency and reduce costs, and as social awareness of the environmental advantages of renewable energy increases.
- 3 (In table order)
 - Energy source in isolated regions
 - Offshore electricity supply and communications
 - Flashing signs and safety lights
 - Lighting and electricity supply in isolated locations
 - Power supply in isolated regions
 - Heating and hot water supplies
 - Use solar panels for power supply
 - Replace batteries with solar power

- Replace batteries
- Where power poles are too expensive to build, e.g. mining towns
- For isolated regions, e.g. bores
- For example, for yachts and isolated locations
- Transportable and convenient, e.g. electricity and heating
- Electricity for lighting and heating in isolated and remote regions of space

4 Advantages: Renewable energy source; Clean; Non-polluting; Individual users can generate their own power without large scale power stations; No moving parts to break; Low maintenance; Limitless energy; Free energy; Flexible to growing needs, e.g. each new roof can have rooftop power; Unlike fossil fuels, does not release carbon dioxide

Disadvantages: Cells are expensive; Cells are inefficient; Needs sunshine to operate; Doesn't work at night time.

5 a Petroleum 37.2%; Natural gas 21.7%; Coal 21.1%; Hydroelectric 6.7%; Nuclear 6.3%; Geothermal, solar, wind and wood 7.0% **b** 7.0% **c** 13.7% **d** 86.3%

6 Pollution, contribution to global warming, not a renewable resource – that is, resources will run out.

10 False, True, False, True, True

Alexander Parkes (pages 12–13)

- 1 plastic, materials, consumption, tonnes, today, 20, carbon, natural, modified, polyethene.
- 2 Answers could include: polyethene – plastic bags, water pipes, squeeze bottles; polypropylene – rope, thermal underwear, polymer bank notes; PVC – magnetic stripe cards, clothing, upholstery; teflon – non-stick coating for cookware, water pipes, lubricants.
- 3 polyethene – hydrogen, carbon; polypropylene – hydrogen, carbon; PVC – hydrogen, carbon, chlorine; teflon – carbon, fluorine
- 6 (In table order) polyvinyl chloride; polyethylene; polyester; polystyrene; bakelite; polypropylene; nylon; polycarbonate
- 8 Thrown away after use, resulting in a huge accumulation of waste
- 9 286
- 11 **Advantages:** versatile; lightweight; durable; resistant to chemicals, water and impact; safe and hygienic; thermal and electrical insulation; inexpensive to produce; some can be recycled; create secondary industries.
Disadvantages: made from non-renewable fossil fuels; can release toxic fumes when burnt; resistant to chemicals, water and impact; generally non-biodegradable; can hurt fauna, e.g. birds and fish; some can be recycled; fill landfill sites
- 12 Uses less energy
- 13 For example, some plastics are reused in making furniture and other useful equipment.
- 16 False, True, True, False, True, True, True

Albert Einstein (pages 14–15)

- 1 a Lead (Pb) b Atoms are unstable c Energy in the bonds that hold the atom together
- 2 b Declining c 5730 years
- 3 One neutron splits a uranium atom into a barium and krypton atom, and this releases large amounts of energy
- 4 Neutron
- 5 Barium and krypton
- 6 Gamma radiation
- 7 The bonds holding the atom together
- 8 Atoms are joined together to form new elements
- 9 Mass is transformed into energy
- 10 Atoms are split in the explosion and energy is released, as seen by the creation of the mushroom cloud

- 11** The clock at the top of the mountain, as it is further from the centre of the Earth
12 Light is bent by gravity
13 **a** False **b** False **c** True **d** True

Anne Jean Macnamara (pages 16–17)

- 1** viral, water, 50, five, bloodstream, nervous, paralysis, die, breathe, iron, polio, vaccine, immunisation
3 Poorer countries, with areas where there is often no clean water and limited sanitation and medical facilities
4 Limited access to clean water, hygienic living conditions, health facilities and vaccinations
5 Poor hygiene and sanitation
7 Incidence of polio has rapidly declined
8 Yes
9 Children are not vaccinated
10 Fear of possible side effects
11 **a** True **b** True **c** True **d** False

Philo Farnsworth (pages 18–19)

- 1** Radio, fibre, light, electronic, cathode, microwave, higher, data
2 Light from the object televised is converted into electrical signals
4 Initially, few people had television sets, but the rate of purchase gradually increased, due to technological developments and mass production lowering costs.
6 (In table order) 16 September 1956, 1993, 1971, 2012, 9 December 1961, 1966, 1960, 20 July 1969, 1 March 1975, 1978, October 1980, 1986, 1994, 1950s, 1 January 2001, 22 November 1956, 15 June 1967
7 Students should point out that Melbourne and Sydney received television services quite quickly, but it took a long time to extend television broadcasting to all of Australia, due to the large distances and low population densities of some areas.
8 Students should realise that buying a television set in the 1950s was a big investment, and that television sets are relatively very cheap today.

Willem Kolff (pages 20–21)

- 1** Two, spine, rib, fist, water, bones, red, pressure
2 Maintaining water balance, blood pressure, mineral levels, pH levels and removal of waste
4 Chronic kidney disease is caused by many factors, including diabetes, high blood pressure, infections, physical injury, old age and hereditary faults. There are also many types of kidney disease.
5 (In table order) kidney cancer, kidney cysts, kidney infections, kidney stone, polycystic kidney.
7 Improves kidney function for a period of time
8 6 years
9 A compatible donor kidney may not be available
10 Filters waste from the blood

Edwin Hubble (pages 22–23)

- 2** Way, solar, stars, spread, shape, Local, spiral, light, 28 000
4 Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune
5 Mercury – 58 million kilometres
6 Jupiter – 143 million kilometres
7 Venus – 243 days
8 Neptune – 165 years
9 The closer a planet is to the Sun, the shorter the year.
10 Pluto, because it does not fit the official definition of a planet, decided on by the International Astronomical Union in 2006.
13 Straight line – shows a direct relationship between the two factors

- 14** From the space shuttle
15 No interference from the atmosphere
16 2.4 metre diameter and 12.3 kg weight
18 The telescope receives more light, so can see greater distances.

Percy Spencer (pages 24–25)

- 1** **a** 10^0 metres **b** No **c** Radio, microwaves, infrared, ultraviolet, X-rays and gamma rays **d** Satellites and mobile phones
3 Quick, simple and efficient
4 Can cause sparks and fires
5 Food can be contaminated by melted plastic
6 It explodes
8 **a** Emergency communication **b** Quick, simple and inexpensive form of communication **c** For emergencies and safety reasons **d** Safety hazard – users do not concentrate fully on driving **e** High debt levels, low recycling rates **f** Possible answers include phones ringing at public events, e.g. theatre; invasion of privacy, e.g. photos taken without knowledge **g** Fear of high levels of electromagnetic radiation
9 Remote, satellites, based, thermal, atmosphere, bushfires, weather, land, water, Antarctica
10 The microwaves are closer together in front of a vehicle and further apart behind it; this can be used to determine the speed of the vehicle.
11 Radar could be used to identify approaching aircraft, ships and other craft
12 **a** False **b** True **c** True **d** False **e** True **f** True

James Watson (pages 26–27)

- 2** goldfish, cattle, chimpanzee, human, cat, cabbage, garden pea, fruitfly
3 Contain few genes, easy to observe results of breeding programs, e.g. flower colour
4 No, goldfish has more chromosomes than humans and chimpanzees, but are much less complex
6 Exponential curve. Frequency of Down Syndrome rapidly increases with age
7 Mutagenic agents such as radiation
8 Engineering, DNA, duplication, embryo, organ, fingerprinting, paternity, insulin, clotting, coding, modified, replacing, diseases
9 Increase quality, yields and disease resistance
11 Cell from udder and unfertilised egg (with nucleus removed) are fused, then placed in a surrogate mother to produce a clone
12 **a** genetic engineering **b** DNA **c** gene **d** chromosome **e** genetics **f** cloning **g** phenotype **h** dominant genes **i** recessive genes **j** transgenic **k** DNA fingerprinting **l** genotype **m** human genome project **n** gene therapy **o** biotechnology **p** genetically modified food **q** gene mapping

Henrietta Leavitt (pages 28–29)

- 1** Bodies in space, consisting mainly of hydrogen and helium and producing light by hydrogen fusion
2 Nuclear fusion
3 In nebulae
5 A small star is produced in an interstellar gas cloud and is converted to a supergiant, white dwarf or black dwarf. A main sequence star is greater than one solar mass and can become a neutron star, pulsar or black hole
8 Hydrogen and helium
9 Nuclear fusion
10 Provides us with energy, light and heat
11 Will eventually explode

Bernard Gordon (pages 30–31)

- 1 CT, X-rays, technology, rotates, images, muscle, osteoporosis, heart
- 3 High intensity where waves are concentrated, and low where waves are less concentrated
- 6 It saves lives and prevents damage if people can prepare for disasters
- 8 **a** transmitter **b** Doppler effect **c** radar **d** CT scanner **e** foetal monitor **f** meteorology **g** military
- 9 **a** True **b** False **c** False **d** True **e** True **f** False **g** True **h** False

Gertrude Elion (pages 32–33)

- 1 Breast, prostate, lung, bowel, liver
- 2 Lung cancer
- 3 Males – lung cancer
- 4 Females – breast cancer
- 5 Smoking
- 6 Prostate and breast cancer
- 7 Sun protection
- 9 Can encourage people to improve their diets, exercise levels and minimise harmful factors such as smoking
- 10 Greek, circulatory, marrow, cells, red, platelets, death, leukaemia, females, genetic
- 12 **a** bone marrow transplants **b** radiation therapy **c** chemotherapy **d** natural remedies
- 13 There is little evidence of significant benefit
- 15 **a** True **b** False **c** True **d** False **e** False **f** True **g** True **h** False

Williard Libby (pages 34–35)

- 1 6 protons and 6 neutrons
- 2 Different number of neutrons, same number of protons
- 6 5730 years
- 8 **a** carbon cycle **b** carbon dating **c** half life **d** organic compounds **e** isotopes
- 9 All reduce the reliability of carbon dating
- 10 **a** False **b** True **c** True **d** True **e** True **f** True

Frank MacFarlane Burnet

(pages 36–37)

- 1 Particle consisting of DNA and protein coat
- 3 Spanish flu
- 4 Death, poverty, physical and emotional suffering
- 5 Poor hygiene and sanitation
- 7 Only multiply inside a living host
- 8 The immune system is not as strong in these groups
- 9 The flu virus mutates (changes)
- 11 So the body has time to develop antibodies, ready to fight flu.
- 12 Minimise contact and transmission
- 13 dollars, two, avian, viral, bird, mortality, faeces, indirect, manure, cooking, humans
- 15 Close contact of many birds
- 16 Loss of income, poverty

Helen Newton Turner (pages 38–39)

- 3 (In table order) fibre diameter, length, soundness, colour, softness, lustre, density
- 4 **a** carpet **b** worsted fabrics, woollen fabrics, and machine knitting **c** hand-knitting yarn, blankets, upholstery and bedding **d** carpets
- 6 Heritability, one, next, percentage, breeding, characteristic, gene, many
- 7 **a** fibre diameter **b** efficiency of weight gain **c** body weight **d** post weaning weight gain

Graeme Clark (pages 40–41)

- 2 Physical sound vibrations are converted into nervous impulses in the ear
- 3 hearing, deafness, globally, population, middle, hereditary, jaundice, age, loud
- 4 Recessive gene
- 5 Has the gene for the condition but does not show or display the condition
- 6 Received the gene from both parents and therefore shows the condition
- 7 All could

Ian Frazer (pages 42–43)

- 1 cancer, virus, cervix, uterus, born, lining, death, teenager, immune, transferred, rate, hundred
- 2 **a** safe sex **b** virus **c** pap smear **d** cervix **e** cervical cancer **f** HPV **g** vaccine
- 6 Cancer rate increases with age
- 8 Developing countries
- 9 Good medical facilities
- 10 Early detection can result in survival rather than death
- 11 As an additional precaution, the vaccine only protects women against certain strains of the human papilloma virus (HPV), which are known to cause 7 out of 10 cases of cervical cancer and 9 out of 10 cases of genital warts.

Crossword of legendary science

(pages 44–45)

Across

- 1 Macnamara; 3 Jeffreys; 7 food chain; 8 radio waves; 10 Parkes; 11 Einstein; 15 Watson; 16 solar cell; 17 Kolff; 21 Leavitt; 23 stars; 24 Hubble; 26 Gordon; 28 Turner; 30 cancer; 32 microwave; 33 cervical; 34 time; 35 objective; 36 bioplastic; 37 Sabin; 38 influenza; 39 radar

Down

- 2 Carson; 4 Clark; 5 Jansky; 6 kidney dialysis; 9 Farnsworth; 12 DNA; 13 pulsar; 14 Ohl; 18 Libby; 19 Spencer; 20 radiocarbon dating; 22 Elion; 25 Burnet; 27 Frazer; 29 engineering; 31 Milky Way; 33 Cochlear

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Jean Macnamara – Polio	Graeme Clark – Bionic ear
Philo Farnsworth – Television	Ian Frazer – Cervical cancer and vaccines
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